

Occupational, Industry, and Geographic Exposure to Artificial Intelligence: A Novel Dataset and Its Potential Uses

Running head: Occupational, Industry, and Geographic Exposure to AI

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Research Summary

We create and validate a new measure of an occupation's exposure to AI that we call the AI Occupational Exposure (AIOE). We use the AIOE to construct a measure of AI exposure at the industry level, which we call the AI Industry Exposure (AIIE) and a measure of AI exposure at the county level, which we call the AI Geographic Exposure (AIGE). We also describe several ways in which the AIOE can be used to create firm level measures of AI exposure. We validate the measures and describe how they can be used in different applications by management, organization and strategy scholars.

Managerial Summary

Although artificial intelligence (AI) promises to spur economic growth, there is widespread concern that it could displace workers, alter industry trajectories, and reshape organizations. Despite the interest in this area, we have limited ability to study the effects of AI on occupations, firms, industries, and geographies because of limited availability of data that measures exposure to AI. To address this limitation, we create and validate a new measure of an occupation's exposure to AI that we call the AI Occupational Exposure (AIOE). We use the AIOE to construct a measure of AI exposure at the industry level (AIIE) and county level (AIGE). We describe how our measures can be useful to scholars and policy-makers interested in identifying the effect of AI on markets.

Keywords: artificial intelligence, industry, innovation, occupation, technology

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1. INTRODUCTION

Recent advances in artificial intelligence (AI) have generated excitement about AI's potential to spur economic growth, and scholars believe that AI has the potential to be “the most important general-purpose technology of our era” (Brynjolfsson & McAfee, 2017). However, there is concern that advances in AI may also have significant consequences for labor markets, firms, and industries by displacing workers, transforming occupational jurisdictions, altering strategy, and affecting performance. For decades now, scholars have considered whether and how rapid advances in information technologies change the nature of competition and strategy (Bennett & Hall, 2019; Bettis & Hitt, 1995; Tippins & Sohi, 2003). In recent years, researchers have increasingly begun to investigate how AI affects firm design, strategy, organizational learning, and management (e.g., Balasubramanian et al., 2020; Bughin et al., 2019; Iansiti & Lakhani, 2020; Jia et al., 2020a, 2020b; Khashabi & Kretschmer, 2019; Raj & Seamans, 2019; Wuebker et al., 2018). However, despite great interest in the academic literature and public press about AI's effects on occupations, firms, and markets, there has been little systematic collection of evidence. Part of the reason for the lack of evidence is that the rapid advancement in AI is a nascent phenomenon, and accordingly, appropriate tools to measure its impact have yet to be developed (McElheran, 2018; Raj & Seamans, 2018).

In order to fill this gap, we develop a new measure of the exposure to AI across occupations that we call the AI Occupational Exposure (AIOE).¹ This measure links common and general applications of AI to workplace abilities and occupations. We show the potential of the occupation-level AIOE by using it to construct two derivative measures. We aggregate the AIOE to the industry level to construct an AI Industry Exposure (AIIE) as well as to the county-

¹ The AIOE measure is a modified version of a method described by Felten, Raj, and Seamans (2018).

level to construct a geographic measure of AI exposure that we call the AI Geographic Exposure (AIGE). We also describe several ways in which the AIOE can be used to create firm level measures of AI exposure. We validate our AIOE measure and then describe how these measures can be used in different applications by strategy, management, and innovation scholars. We have made these datasets freely available for use by scholars, policy-makers, and practitioners.²

We build our measures by linking common AI applications to occupational abilities using a crowd-sourced dataset. We aggregate the effect at the ability level to construct a measure that identifies the potential exposure of occupations to AI. While similar in spirit to recent work (e.g., Brynjolfsson et al., 2018; Frey & Osborne, 2017; Georgieff & Milanez, 2021; Tolan et al., 2020; Webb, 2020), our methodology is unique in that it links specific applications of AI to occupational abilities and is agnostic as to whether AI substitutes for or complements workplace abilities and occupations. We discuss differences with existing datasets in more detail below. We believe our methodology has the potential to be used in a variety of applications within the field of strategy and, as we describe below, can be used to construct AI exposure at the firm level as well (see, e.g., Acemoglu et al., 2020).

Our paper contributes in several ways. First, we develop a new methodology linking applications of AI technology to human abilities. We use this method to generate occupation-level, industry-level, and geographic-level “AI Exposure” measures, which are available for other researchers to use. By describing the data construction and validation in the body of this paper and providing this data for use by other researchers, we are answering a call from Ethiraj et al. (2019) for more articles that develop and describe datasets for other researchers to use. Second, we situate our measures within existing data by comparing the AIOE with other

² The datasets can be accessed via GitHub: <https://github.com/AIOE-Data/AIOE>.

measures that link work and occupations to AI, automation, and robotics (e.g., Brynjolfsson et al., 2020; Frey & Osborne, 2017; Webb, 2020). Third, we describe several ways in which the measures can be used by researchers to study the effects of exposure to AI on workers, firms, industries, and geographies.

The paper proceeds as follows. In the next section, we outline the methodology that we use to construct the AI Occupational Exposure (AIOE) and its accompanying AI Industry Exposure (AIIE) and AI Geographic Exposure (AIGE) measures. In the third section, we describe the steps we take to validate our methodology and resulting measures. In the fourth section, we describe potential applications for the datasets, with particular attention to the use of the AIOE for organizational level studies. The fifth section concludes.

2. CONSTRUCTION OF THE AI EXPOSURE MEASURES

Artificial intelligence is a construct with varying definitions and interpretations: “general” AI refers to computer software that can think and act on its own, which does not yet exist; “narrow” AI refers to computer software that relies on highly sophisticated algorithmic techniques to find patterns in data and make predictions about the future (Broussard, 2018). Because narrow AI algorithms “learn” from existing data to improve performance, these techniques are often referred to as “machine learning.” To construct our measure of occupational exposure to AI, we start by identifying different applications of these machine learning techniques and then link these applications to occupational abilities.

Our methodology isolates the exposure to specific functions of AI and creates a measure of the aggregate exposure to AI at the occupation level by looking at the abilities used within an occupation. We consider “labor” as the bundle of skills and abilities that are used within a specific occupation (D. Autor & Handel, 2013; Cohen, 2012), and this “micro” approach allows

us to examine how AI may be related to the abilities that each occupation requires. We link specific functions of AI to different types of abilities in the US workforce using a crowdsourced dataset; then, considering the work content of each occupation, we aggregate the ability-level exposure to AI applications at the occupation level. Note that our measure of exposure does not attempt to measure whether AI is a complement to or a substitute for labor, but rather how likely it is that the occupation is exposed to AI in some way.

Our analytical strategy borrows in part from Felten et al.'s (2018) methodology, which links progress in AI applications to construct a backward-looking measure that evaluates the effect of AI on occupations from 2010 to 2015. We focus on a similar set of AI applications to build our measure, but instead of accounting for historical progress to construct a backward-looking measure, we take a forward-looking approach to identify occupational exposure to AI. We also modify Felten et al.'s (2018) methodology to scale the aggregate exposure to AI at the occupation level by the portfolio of abilities used within an occupation. This modification allows us to account for occupation scope and more accurately measure the relative occupation-level exposure to AI.

2.1. Data

To construct our measure, we identify a set of common and well-developed applications of AI based on categories defined and described by the Electronic Frontier Foundation (EFF) AI Progress Measurement project and connect these applications of AI to data on occupational abilities from the Occupational Information Network (O*NET) database developed by the United States Department of Labor.

The Electronic Frontier Foundation (EFF) is a well-regarded digital rights non-profit that was founded in 1990, focusing on issues related to digital rights and privacy. Searches in Google

Scholar show thousands of results, including articles written by the EFF as well as articles that cite EFF's work in a variety of fields, including law, economics, technology, and others (e.g., Casadesus-Masanell & Hervas-Drane, 2015; Eaton et al., 2015; Farrell & Shapiro, 2008; Liebowitz, 2006; Lu et al., 2019).

As one of its activities, the EFF collects and maintains statistics about the progress of AI across separate artificial intelligence applications. For each application, the EFF monitors progress in the field by drawing on data from multiple sources, including academic literature, review articles, blog posts, and websites focused on artificial intelligence. For all information collected, the EFF only includes data from verified sources that document proof of their findings (*AI Progress Measurement*, 2019), and the EFF AI metrics have been cited in various computer science outlets (e.g., Carlson, 2019; Croeser & Eckersley, 2019; Krinitskiy et al., 2018). For the purposes of this study, we focus on the ten AI applications for which the EFF has recorded measured scientific activity and progress in the technology from 2010 onward as we believe these are the applications experiencing the fastest growth and most likely to be used in the medium-term.³ While we use the EFF categorization to identify applications of AI, we note that EFF's categorization is consistent with the categorization of AI applications in other outlets (e.g., Bessen et al., 2018; Martínez-Plumed et al., 2020; Perrault et al., 2019).

The O*NET data define and describe professions in the modern-day American workplace (*O*NET® Database Releases Archive at O*NET Resource Center*, 2019) and are frequently used to measure occupational work or task content in academic research (e.g., D. Autor &

³There has been less activity in the EFF AI Progress Measurement project since 2017. However, we believe that the ten applications we have selected are appropriate representations of common applications of AI technology. We discuss how the EFF categorizations compare to other sources and explore the robustness of the measure to using a subset of applications in *Online Appendix C*.

Handel, 2013; Brynjolfsson et al., 2018; Goos et al., 2009). For the purposes of our analysis, we focus on the 52 distinct abilities O*NET uses to describe the workplace activities of each occupation. O*NET notes both the importance and the level or prevalence of each ability within an occupation (using a 1–5 and 1–7 scale respectively).

2.2 Methodology

2.2.1 Selection of AI Applications

We construct the AIOE using data from ten selected AI applications: abstract strategy games, real-time video games, image recognition, visual question answering, image generation, reading comprehension, language modeling, translation, speech recognition, and instrumental track recognition. This set of applications does not comprehensively cover the set of applications for which AI could ultimately be used; however, based on our conversations with field experts, we believe that these represent fundamental applications of AI that are likely to have implications for the workforce and are applications that cover the most likely and most common uses of AI.⁴ Table 1 presents the definition of each application provided by the EFF.

TABLE 1 HERE

2.2.2 Linking AI Applications to O*NET Abilities

To link the AI applications to workplace abilities, for each of the AI applications considered in our analysis (abstract strategy games, real-time video games, image recognition, visual question answering, image generation, reading comprehension, language modeling, translation, and speech recognition), we use a crowd-sourced data set constructed using survey responses of “gig workers” from Amazon’s Mechanical Turk (mTurk) web service. Through this survey, we generate a measure of application-ability relatedness for each combination bound

⁴ Robustness checks around the selection of AI applications are presented in *Online Appendix C*.

between 0 and 1 (survey methodology and measure construction is discussed in more detail in *Online Appendix A*). We organize this measure of application-ability relatedness into a matrix that connects the ten EFF AI applications to the 52 O*NET occupational abilities. Using the measures of application-ability relatedness from this matrix, we then calculate an ability-level exposure as follows:

$$A_{ij} = \sum_{i=1}^{10} x_{ij} \quad (1)$$

In this equation, i indexes the AI application and j indexes the occupational ability. The ability-level exposure, A , is calculated as the sum of the ten application-ability relatedness scores, x , as constructed using mTurk survey data. For example, if the application-ability relatedness score between an occupational ability and each separate AI application were 0.5, the ability-level exposure would be 5.0. By calculating the ability-level AI exposure as a sum of all the AI applications, we are equally weighting each application. Some may argue with this choice, especially given that certain applications appear related (e.g., visual question answering and image recognition). Although we choose this approach for simplicity and to avoid making arbitrary decisions about whether and how to weight applications, we show in *Online Appendix C* that our measures are largely consistent if we construct our AIOE measure using alternative sets of applications, suggesting that weighting the applications is unlikely to have a meaningful impact on the measure.⁵ Our approach also assumes that each application has an independent effect on an ability and does not consider interactions across applications. Although this is a limitation, considering the myriad ways in which the applications could interact would be infeasible.

2.2.3 Calculating Occupational Exposure

⁵ This is because we find that, across the different applications of AI, the relationship between AI and abilities is strongest for “cognitive” abilities.

We next use the O*NET occupational definitions to evaluate the exposure to the AI technology for each occupation. We rely on the O*NET 24.3 database released in May 2020, as it is the most up-to-date version of the O*NET data at the time of analysis. Our occupation-level AIOE measure is constructed as follows:

$$AIOE_k = \frac{\sum_{j=1}^{52} A_{ij} \times L_{jk} \times I_{jk}}{\sum_{j=1}^{52} L_{jk} \times I_{jk}} \quad (2)$$

In this equation, i indexes the AI application, j indexes the occupational ability, and k indexes the occupation. A_{ij} represents the ability-level exposure score calculated in equation 1. We weight the ability-level AI exposure by the ability's prevalence (L_{jk}) and importance (I_{jk}) within each occupation as measured by O*NET by multiplying the ability-level AI exposure by the prevalence and importance scores for that ability within each occupation, scaled so that they are equally weighted.⁶ These prevalence and importance scores allow us to properly account for the presence of different abilities within an occupation. Abilities that are integral to an occupation have high prevalence and importance scores, while those that are used less often or are less vital have lower prevalence and importance scores. For example, for CEOs, O*NET considers abilities such as oral comprehension (importance: 4.5; prevalence: 4.88) and oral expression (importance: 4.38; prevalence: 5) as highly important, and abilities such as speed of limb movement and static strength as non-essential (for both, importance: 1; prevalence: 0).⁷

We measure an occupation's aggregate exposure to AI by summing this weighted ability-level AI exposure across all abilities in an occupation. In an adjustment to Felten et al.'s (2018) methodology, we scale the aggregated exposure to AI across all abilities by the weighted sum of

⁶ To equally weight prevalence and importance, we divide each score by the maximum value O*NET considers for the metric.

⁷ There is a high correlation between an occupation's prevalence and importance scores. The AIOE measure is largely consistent using either prevalence or importance alone to weight the ability-level AI exposure.

the prevalence and importance of all abilities used in the occupation to account for the total required ability set within an occupation. This scaling provides us with a measure of the *relative exposure* to AI. We believe this adjustment is important owing to the nature of the O*NET occupational definitions. Some occupations as defined by O*NET have many abilities that are considered important and prevalent, while others have a smaller number. Because an occupation's *aggregate exposure* to AI is measured by summing across abilities, occupations that use more abilities are likelier to have higher levels of aggregate exposure. Without the adjustment based on the weighted sum of all abilities, our methodology would overweight the exposure to AI for broader occupations that require more abilities.

To clarify this with an example, consider two separate occupations — surgeons and physicists. Both occupations score highly on *aggregate exposure* to AI (calculated by summing the ability-level AI exposure across all abilities). Using the outlined approach, surgeons have the third highest aggregate AI exposure in our sample (3.28 standard deviations above the mean), while physicists have the fourth highest aggregate AI exposure in our sample (3.14 standard deviations above the mean). However, although the aggregate exposure of AI may be similar for these two occupations, the *relative exposure* to AI within the occupation differs greatly because each of these occupations rely on a different portfolio of abilities. In particular, surgeons rely on a broad range of abilities from all ability families — cognitive, physical, psychomotor, and sensory abilities — while physicists rely heavily on a smaller set of largely cognitive abilities. To account for such differences across occupations, we scale the aggregate exposure of AI across an occupation by the breadth of abilities required in that occupation. After that adjustment, we find that physicists are still considered highly exposed to AI (the 61st ranked occupation with an AIOE measure 1.35 standard deviations above the mean), while surgeons are no longer classified

as highly exposed (the 387th ranked occupation with an AIOE measure of 0.05 standard deviations below the mean). The adjustment, therefore, accounts for the different scope of occupations and more accurately measures the relative exposure to AI within an occupation.

The O*NET data are organized according to each occupation's eight-digit SOC classification; we collapse the AIOE to occupations at the six-digit SOC level to align with other major data sources. Because the vast majority of eight-digit SOC classifications contain only one six-digit SOC classification, this change does not alter our sample size greatly. We collapse the 832 occupations at the eight-digit SOC level to 774 occupations at the six-digit SOC level by calculating the mean AIOE across all occupations defined by O*NET within the six-digit SOC level. Finally, we standardize our AIOE measures such that the mean across occupations is zero and the standard deviation is one.⁸ Ultimately, our methodology produces a score that measures how the most prevalent applications of AI are related to occupations at the six-digit SOC level based on their ability composition as defined by O*NET. We again emphasize that this does not constitute a measure for how substitutable or automatable an occupation is, as we are agnostic about whether or when AI will augment or replace human labor.

2.2.4 AIIE and AIGE

We construct our industry-level measure of AI exposure by aggregating the AIOE across occupations within an industry. We construct an AI Industry Exposure (AIIE) by taking a weighted average of the AIOE using industry employment based on the four-digit NAICS classification in 2019, the latest data available at the time of writing. This measure allows us to track which industries have the most exposure to AI. We construct our geographic-level measure of AI exposure by aggregating the AIIE across industries within a geographic area using county-

⁸ This step results in occupations with negative AIOE measures. We do not intend to suggest that occupations can have negative exposure to AI; rather this transformation just allows for ease of comparison across occupations.

level employment in 2019 (we use the US county designations (FIPS codes) to construct measures at the county and state levels). In addition to providing us with a measure of AI exposure at industry and geographic levels, this exercise demonstrates the flexibility of our methodology. Our measures can be updated based on the changing occupational descriptions and definitions provided by the O*NET database, changing occupational composition within an industry, or changing occupational composition within a geographic area. In addition, we believe that this measure can be aggregated to other levels, such as the firm or subindustry level, using the distribution of employment or wages across occupations within a grouping.

We provide the datasets created by our methodology in the *Data Appendices* available via GitHub.⁹ A list of occupations and the associated AIOE measures is in *Data Appendix A*; a list of industries at the four-digit NAICS classification measuring industry exposure to AI technology is in *Data Appendix B*; and the AIGE scores, measuring geographic exposure to AI technology, are in *Data Appendix C*. We include the matrix connecting AI applications to occupational abilities in *Data Appendix D* and the standardized ability-level AI exposure in *Data Appendix E*. In the following section, we analyze and validate the AIOE measures.

3. VALIDATION OF THE AI OCCUPATION EXPOSURE SCORES

Owing to the novel nature of our methodology and the limited research in this area, validation of the metric presents challenges since there are no clear benchmarks to compare with the AIOE measures. It is unclear how one might expect the exposure to AI to manifest itself in occupations while remaining agnostic about its effect on occupations and patterns of adoption, and most existing metrics for measuring occupation-level exposure to AI seek to determine the likelihood of substitution (Brynjolfsson et al., 2018; Frey & Osborne, 2017).

⁹ The datasets can be accessed via GitHub: <https://github.com/AIOE-Data/AIOE>.

Given these challenges, we validate the AIOE measures in several ways. First, as an initial “gut check,” we qualitatively discuss the differences across occupations that receive the highest and lowest scores in our sample. We show that exposure to AI seems to be highest in white-collar occupations and industries. Second, we provide a deeper discussion for several occupations, including some that are often considered to be threatened by advances in AI such as taxi drivers and long-haul truck drivers, to provide a sense of how advances in AI are related to specific occupational abilities. Third, we present robustness checks for the construction of our matrix that connects AI applications to abilities and show that our scores are mostly consistent even if we exclude different applications of AI. Fourth, we use data from job postings to present evidence that the AIOE is positively associated with the use of AI skills within an occupation.¹⁰

3.1 Most- and Least-Exposed Occupations and Industries

To begin to understand the nature of the AIOE measure, we first compare the highest-scoring occupations with the lowest-scoring occupations. These represent the occupations with the most and least reliance on abilities that are likely to be affected by advances in AI technology. Table 2 presents the twenty occupations with the highest and lowest AIOE measures in our sample.

TABLE 2 HERE

The highest-scoring occupations (i.e., the ones that the AIOE predicts are most exposed to advances in AI technologies) consist almost entirely of white-collar occupations that require advanced degrees, such as genetic counselors, financial examiners, and actuaries. The lowest-scoring occupations are largely non-office jobs that require a high degree of physical effort and exertion, such as dancers, fitness trainers and aerobics instructors, and painters and plasterers.

¹⁰ While we present the highest- and lowest-scoring industries based on AIIE, we largely focus on validating our AIOE measure rather than the AIIE measure since the AIIE is constructed using the AIOE.

We can similarly compare the highest- and lowest-scoring industries based on our AIIE measure to identify which industries our methodology suggests have the most exposure to AI technologies. Table 3 presents the twenty industries with the highest- and lowest-scoring AIIE measure in our sample.

TABLE 3 HERE

Just as with the ranked occupations, we find that the most exposed industries tend to be white-collar industries requiring high levels of education, such as financial services, accounting, insurance, and legal services. Perhaps not surprisingly, exposure to AI is particularly high in certain service industries that involve a high level of information processing. On the other hand, the lowest-scoring industries tend to be blue-collar industries that involve manual labor, such as support activities for crop production, building and dwellings services, construction contracting services, and warehousing and storage. Intuitively, these occupation and industry rankings reflect 1) our aim to isolate the exposure to advances in AI (as opposed to, say, robotics, machine vision, autonomous guided vehicles, or other types of advanced technologies), and 2) the abilities most likely to be affected by AI. Whereas robotic technologies often involve physical manipulation and are capable of performing complex manual tasks, AI technologies are largely software-based and rely on iterative learning and perception (Raj & Seamans, 2019). Accordingly, we expect AI to have a limited influence on the role of physical abilities in occupations and industries. Instead, our methodology leads us to believe that AI is likely to have the biggest impact on abilities related to information processing. The AIOE and AIIE measures reflect the larger influence of AI on cognitive tasks and abilities and a smaller influence on physical tasks and abilities. We believe that this relationship is in line with current scientific

perceptions of AI technologies (e.g., Brynjolfsson & McAfee, 2014; Brynjolfsson & Mitchell, 2017) and accordingly provides a measure of support for our methodology.

3.2 Discussion of Selected Occupations

We next turn to a discussion of selected occupations. We start by comparing and contrasting the scores of two occupations that require some similar abilities but have disparate AI exposure scores — surgeons and meat slaughterers — to highlight the importance of cognitive abilities in determining occupational exposure to AI. We then compare two occupations that are similar in the presence of cognitive abilities but have dissimilar AOIE scores — mathematical technicians and accountants and auditors — to examine what other facets of an occupation may affect exposure to AI. In *Online Appendix B*, we also review two occupations frequently discussed in the context of AI — truck drivers and taxi drivers.

3.2.1 Surgeons and Slaughterers

We attempt to provide more context for the AIOE measures by comparing two professions that share some similarities abilities required yet have highly disparate AIOE measures — surgeons and meat slaughterers. Superficially, we might expect these occupations to have similar AIOE measures, as both require deft physical manipulation of human or animal tissue. Although the occupations require similar physical abilities, such as manual dexterity, finger dexterity, and arm-hand steadiness, the occupations' AIOE measures suggest that surgeons are far more exposed to AI than slaughterers. The AIOE measure for surgeons is at the 52nd percentile in relation to other occupations in our sample, while the AIOE measure for slaughterers is at the 2nd percentile (indeed, it is the 10th least-exposed occupation).

The difference between the AIOE measures in these two occupations seems to arise from the cognitive abilities required by each occupation. Although the two occupations require similar

physical abilities, a number of cognitive abilities related to problem solving, such as problem sensitivity, deductive and inductive reasoning, and information ordering, are highly important for surgeons but not for meat slaughterers. Our methodology suggests that occupations' exposure to AI largely stems from these cognitive abilities; accordingly, surgeons have a much higher AIOE measure than meat slaughterers.

Generalizing from this specific example, our AIOE measure emphasizes that the presence of cognitive abilities plays a large role in determining how exposed an occupation is to AI. This has implications for how work content and processes may be affected by advances in AI. Literature has argued that computer-based technologies may have a large effect on cognitive tasks, substituting for routine cognitive tasks and complementing nonroutine cognitive tasks (D. H. Autor et al., 2003). The AIOE measure suggests that, like previous iterations of computer-based technologies, AI is likely to disproportionately affect cognitive tasks and occupations. We expect occupations that require a greater amount of problem solving, logical reasoning, and perception to be more exposed to AI than occupations that largely require physical abilities.

3.2.2 Mathematical Technicians and Accountants & Auditors

As discussed above, we find that the primary driver of the AIOE is the presence of cognitive abilities within an occupation. However, because our methodology relies on granular abilities, we can look beyond just the presence of cognitive abilities to examine what other occupational characteristics are related to exposure to AI. To do this, we consider two occupations that require a similar level of cognitive abilities but have dissimilar AIOE scores – mathematical technicians and accountants and auditors.

Based on O*NET's occupational definitions and categorization of abilities, 80.0% of abilities required for both these occupations (weighted by importance and level) are cognitive

abilities. These are both occupations that could be considered relatively “cognitive”, as for the median occupation, 56.8% of abilities required are cognitive abilities. However, despite a similar level of cognitive abilities in these two occupations, there is a disparity in the AIOE scores.

Accountants and auditors have an AIOE score at the 99th percentile relative to other occupations in our sample, while mathematical technicians have an AIOE score at the 68th percentile.

The difference is accounted for by the presence of sensory vs. physical or psychomotor abilities. Cognitive abilities are more exposed to AI relative to all other abilities; however, differences remain across non-cognitive abilities. In particular, we find that sensory abilities, defined as abilities that influence visual, auditory, and speech perception, are more exposed to AI technologies than physical or psychomotor abilities. This is perhaps not surprising given that AI technologies are particularly known to be well-suited for tasks involving classification, categorization, and pattern recognition (Broussard, 2018; Choudhury et al., 2020; Kotsiantis et al., 2006), and perception may be keenly involved in such tasks. Of the non-cognitive abilities required for accountants and auditors, 90.6% are sensory abilities, while for mathematical technicians, only 52.6% are sensory abilities. The higher relative importance of sensory abilities (vs. physical or psychomotor abilities) for accountants and auditors results in the higher exposure to AI relative to mathematical technicians, despite a similar level of cognitive abilities.

3.3 Quantitative Validation

In addition to our qualitative review of occupations and their AIOE measures, we test the robustness of our AIOE measure and seek to validate the measure quantitatively. We describe these tests briefly here and discuss them in more detail in *Online Appendix C*. We test the robustness of our measure along four separate dimensions: (1) we consider the construction of our matrix linking AI applications to occupational abilities and show that the matrix connecting

AI applications and O*NET abilities remains consistent using a subset of respondents with graduate degrees or with computer science or engineering degrees; (2) we test the sensitivity of the measure to which set of AI applications is included and show that our measure is robust to alternative application selection, as all AI applications are most related to cognitive and sensory abilities; (3) we use job postings to show a strong positive correlation between the AIOE measure and the use of AI skills within an occupation; and (4) we examine how changes in occupational definitions by O*NET affect the AIOE measure and show that the measure is relatively consistent over short- to medium-term time horizons.

4. POTENTIAL APPLICATIONS OF AIOE, AIIE, AND AIGE

We believe our AIOE, AIIE, and AIGE measures can be used by a range of scholars interested in occupational, industry, firm, and regional outcomes. Thus, our measure should be useful to scholars investigating research questions in multiple areas including competitive and corporate strategy, organizational design, human resource management, industrial organization, geography of innovation, regional and business dynamism, and entrepreneurship. In this section, we describe how the AIOE, AIIE, and AIGE measures can be used by scholars to better understand how exposure to AI is affecting occupations, firms, industries, and regions. We first describe distinctions between the measures and existing datasets at a high-level to inform of particularly suitable uses of the data before providing some examples of research questions that the data could be used to examine.

4.1 Distinctions between the AIOE and Existing Datasets

The AIOE and related measure are distinct from existing datasets in a number of ways. First, our methodology considers specific applications of AI (e.g., image recognition, speech recognition, and others) and links them to workplace abilities and then to occupations, industries,

and geographies, rather than considering the effect of automation (Frey & Osborne, 2017; Mann & Püttmann, 2017) or AI more broadly (Brynjolfsson et al., 2018, 2020; Webb, 2020) on labor, without specifying whether automation occurs via AI, robots, sensors, or another type of technology. Second, our measure identifies the relative *exposure* to AI but remains agnostic as to whether AI complements or substitutes for tasks and labor (e.g., Brynjolfsson et al., 2020; Frey & Osborne, 2017; Webb, 2020). Third, our approach provides a snapshot of occupational exposure to AI based on the current nature of occupations, rather than relying on expert projections or crowdsourced evaluations to identify which tasks may be suitable to AI or automation or how the task composition of an occupation may change as it is affected by AI (Brynjolfsson et al., 2018, 2020; Frey & Osborne, 2017).

Such distinctions inform us of particularly suitable uses of the data. Given that our measure considers AI (and applications of AI) specifically, we believe our measure is best suited for studies that focus on AI exclusively rather than automation or robotics more broadly. Because our measure seeks to measure exposure to AI rather than identify abilities or occupations that are likely to be replaced or automated, we also believe that our measure can be useful for scholars who aim to explore the trade-off between substitution and complementarities, as it does not seek to identify abilities or occupations that are likely to be replaced or automated. Further, because the AIOE, AIIE, and AIGE are forward-looking measures, we believe they can inform us which occupations, industries, and geographies are most likely to be exposed to AI based on current occupational definitions

4.2 Potential Uses of the AIOE, AIIE, and AIGE Across Fields

4.2.1 Labor Economics

We believe that our AIOE measure can be a valuable tool for scholars seeking to study how advances in AI relate to individual- and occupation-level outcomes within firms. Articles in the public press frequently discuss the threat that developments in AI may pose to workers (e.g., Jordan, 2018; Stark, 2017; Wacker, 2017), and a stream of research has begun to examine how AI and automation may affect the workforce (e.g., Acemoglu & Restrepo, 2018; Arntz et al., 2016; Frey & Osborne, 2017; Mann & Püttmann, 2017). However, despite such interest in the press and in the academic community, there is very little systematic evidence on the impact of AI on labor. Because the AIOE measure is constructed to be agnostic to the effect of AI on labor demand, we believe that the AIOE can be a useful tool for understanding the circumstances in which AI substitutes for or complements labor.

Beyond this fundamental question regarding labor substitution, the AIOE can be used to study the relationship between AI, human capital, and labor productivity. For example, previous literature has suggested that equipment automation does not substitute for the value of human capital, at least within the semiconductor industry (Hatch & Dyer, 2004), but it is unclear whether and how the emergence of AI will alter the value of human capital within firms. While capital-labor substitution has traditionally been driven by highly specialized technologies (Istvan, 1992), AI is a general-purpose technology (Goldfarb, Taska, and Teodoridis, 2019) with a wide range of specialized applications that could challenge this common belief. Our measures may be useful to scholars seeking to extend research seeking to understand how AI and other digital or automating technologies may affect human productivity (Brynjolfsson & McElheran, 2016; Choudhury et al., 2020; Cowgill, 2019; Furman & Teodoridis, 2020).

4.2.2 Organizational Design

The rapid emergence of AI is likely to alter the distribution of value created by employees across an organization (Brynjolfsson & McAfee, 2017; Bughin et al., 2019; Fountaine et al., 2019; Iansiti & Lakhani, 2020). Previous research has examined how ICTs have affected firm boundaries and transaction costs (McElheran & Forman, 2019; Vakili & Kaplan, 2019). Similarly, scholars could use the AIOE to study the links between AI and occupational characteristics and work content, to better understand the implications of AI for organizational design and structure. For example, Raj and Seamans (2019) call for more studies on how AI changes the nature of work and the resulting implications for organizational design, structure, and boundaries.

4.2.3 Competitive and Corporate Strategy

The AIOE can also be used to construct a measure of AI exposure at the firm-level. To do this, a researcher would need the occupational breakdown of employees within firms. The researcher could use the occupation names or codes to link our AIOE measures to the occupations in the firm, and then created an AI exposure score for the firm by weighting the AIOE measure for each occupation present in the firm by the percentage of the firm's employees in that occupation. Several existing datasets allow for this kind of firm-level data construction. For example, the Occupational Employment Survey (OES), which is produced by the BLS, provides publicly available occupational data at the industry level. The underlying micro-data, at the firm level, are confidential but available to BLS-approved researchers, and the OES micro-data have been used to study occupational differences across firms by Ayyagari and Maksimovic (2017), Handwerker (2020) and Ma et al. (2019). As another example, data from the Equal Employment Opportunity Commission (EEOC) allow researchers to study occupational-level details within firms (e.g., Ferguson & Koning, 2018; Koning & Ferguson, 2019; Tomaskovic-

Devey et al., 2016). Finally, the Burning Glass Technologies (BG) data that we use in one of our validation exercises can be used to study occupational differences across firms. Indeed, Acemoglu et al. (2020) match our AIOE measure to firm level BG data to study how a firm's AI exposure affects its hiring of AI skilled and non-AI skilled workers. Interestingly, Acemoglu et al. also do the same for the Brynjolfsson et al. (2018) and Webb (2020) measures and discuss the differences across the three measures.

A firm-level measure of AI exposure could be used to answer a number of questions related to competitive strategy. Strategy and management scholars have often sought to identify the different firm characteristics that predict novel technology adoption or diffusion (e.g., Brynjolfsson & McElheran, 2016; Greve, 2009; Majumdar & Venkataraman, 1998; Souder et al., 2017). We believe that a firm-level construction of our AI measure could be used to identify what characteristics predict firm-level adoption of AI technologies given similar exposure. In a recent article, Adner, Puranam, and Zhu (2019) call for additional research on digital innovation and its implications for firm strategy. A burgeoning line of research has already begun to study how AI affects internal organization and strategy (e.g., Bughin et al., 2019; Iansiti & Lakhani, 2020; Jia et al., 2020a; Khashabi & Kretschmer, 2019). We believe that our measures could be used by such scholars to identify firms that are facing greater exposure to novel AI technologies and study changes in organization or strategy.

4.2.4 Industry Evolution

Technological change can have meaningful consequences for industry structure and lead to the entry or exit of firms within an industry (e.g., Abernathy & Utterback, 1978; Agarwal & Gort, 2002; Klepper, 1996, 1997). Historically, product innovation has led to industry shakeouts and changes in firm innovative strategy (Klepper & Simons, 2000; Utterback, 1987), and already

evidence has emerged that AI has affected talent recruitment, healthcare, and legal industries (Cowgill, 2019; Galasso & Luo, 2018; Ramesh et al., 2004). Literature has suggested that the increased prevalence of ICTs may decrease competition and increase concentration within industries (D. Autor et al., 2017; Bessen, 2017; Van Reenen, 2018). Recent work has drawn links between high-technology sectors and declining business dynamism and has suggested that the rise of ICTs may be partially responsible for declining productivity growth (Bijnens & Konings, 2018; Haltiwanger et al., 2014). Other work suggests, however, that an increase in exposure to software availability increases industry entry and also increases exit by the oldest and most established firms in the industry (Bennett & Hall, 2019). As AI technology continues to advance and diffuse, understanding how the AI affects industry growth and evolution becomes increasingly important. For example, Babina et al. (2020) study how investments in AI affect industry dynamics and find that increased adoption of AI technologies is associated with an increase in industry concentration. We believe the AIIE measure could be used by researchers to study the links between AI and industry concentration, firm entry, or business dynamism.

4.2.5 Economic Geography

Finally, we believe the AIGE measure can be used to study the relationship between AI and regional level outcomes. Previous research has found that technology spillovers tend to be geographically localized because transfer and diffusion of (often tacit) knowledge benefits from local interactions, whether it be through communication, collaboration, or localized human capital (Almeida & Kogut, 1999; Audretsch & Feldman, 1996; Boschma, 2005). The extent to which local interactions occur and whether they will eventually result in meaningful advances in technology may depend upon regional capabilities that govern the innovation processes. These capabilities could include infrastructure, natural resources, institutions, universities, firms, skills,

and culture (Cooke, 2001; Maskell & Malmberg, 1999). For example, local infrastructure such as roads and railways can enhance regional innovation by increasing diffusion of technological knowledge among local inventors (Agrawal et al., 2017; Perlman, 2015). It is possible that a region's exposure to AI generates tacit knowledge about how to work with and benefit from the technology that in turn constitutes a capability upon which future innovation will build. Moreover, one might expect that the effect is greater in areas where there are more interactions, perhaps because of better roads and infrastructure or more foot traffic (Roche, 2019). On the other hand, to the extent that AI substitutes for labor, on net, then a region's exposure to AI will indicate future job losses and economic decline.

As a simple example of the potential of the AIGE, in Figure 1 we present a map of the continental United States that displays the AIGE by county. More highly exposed counties (i.e., those with relatively more employment in occupations with high AIOE measures) are shown in red and less exposed counties in blue.

FIGURE 1 HERE

Figure 1 presents some immediate patterns for AI exposure across the US. It appears that, on average, urban counties are more highly exposed to AI than rural counties. For example, the Atlantic corridor, comprising Boston, New York City, Philadelphia, Baltimore, and Washington, D.C., has a relatively high-level of AI exposure, as does the Bay Area in California. While rural areas have lower AI exposure on average, there are some exceptions. For example, the state of Iowa has a number of counties with relatively high exposure to AI despite being predominantly rural. We present this figure as a simple example of the potential of the data and leave it to future scholars to disentangle the underlying causes and effects of such heterogeneity in exposure.

5. CONCLUSION

Despite excitement about AI's benefits to organizations and concern about its potential effect on labor, there is little systematic evidence of either. Our study makes several contributions toward understanding these effects. First, we describe and validate a new methodology for linking advances in AI to human abilities. Second, using this methodology, we develop a measure of an occupation's exposure to AI, which we call the AI Occupational Exposure (AIOE); a measure of an industry's exposure to AI, which we call the AI Industry Exposure (AIIE); and a measure of a geographic area's exposure to AI, which we call the AI Geographic Exposure (AIGE). We also discuss ways in which our AIOE measure can be aggregated to the firm level. We expect the new measures will be useful to researchers interested in the link between AI and occupation, firm, industry, and region-oriented outcomes. Third, we validate the constructed measures and discuss their potential uses.

We believe that these datasets can be useful to policymakers and scholars seeking to understand how AI will affect individuals, firms, and markets. The methodology used to construct the data is dynamic and can be updated as O*NET updates occupational definitions over time or occupational composition changes across industries and geographies. The AIOE, AIIE, and AIGE can potentially be used by policymakers to identify the occupations, industries, and geographic areas that are most likely to be affected by future advances. These measures can be used by scholars in economics, management, and strategy to answer questions regarding the effect of AI on firm, market, and local outcomes.

Our paper is closely related to a small number of others that provide their own measures of the exposure to new technologies on occupations and industries (Brynjolfsson et al., 2018, 2020; Frey & Osborne, 2017; Tolan et al., 2020; Webb, 2020). We believe that our measures are

unique in that they focus on specific applications of AI technologies rather than considering AI or automation broadly; they are agnostic as to whether the effect of AI serves as a complement or a substitute; and they provide a snapshot of occupational exposure to AI based on the current nature of the occupation. We believe our approach complements existing approaches and can be useful to researchers, depending on the specific questions being asked. Some researchers have already used the AIOE, including Fossen and Sorgner (2019), who study how advances in AI create opportunities for growth-oriented entrepreneurship, Goldfarb et al. (2019), who use the AIOE to examine whether AI has the characteristics of a general-purpose technology within the healthcare industry, and Acemoglu et al. (2020) who use the AIOE to study how a firm's AI exposure affects hiring patterns. It is our hope that future work can continue to investigate how the exposure to AI manifests across occupations, geographies, and backgrounds.

AI technologies are rapidly advancing and are expected to have dramatic effects on the economy. Given the sizable effect that we suspect AI will have on firms and markets, it is critical to have measures that allow us to study this important phenomenon. While much work must still be done to examine the relationship between AI and labor, we believe our measures can be useful tools in this endeavor and that our study contributes to the growing body of literature examining this important topic.

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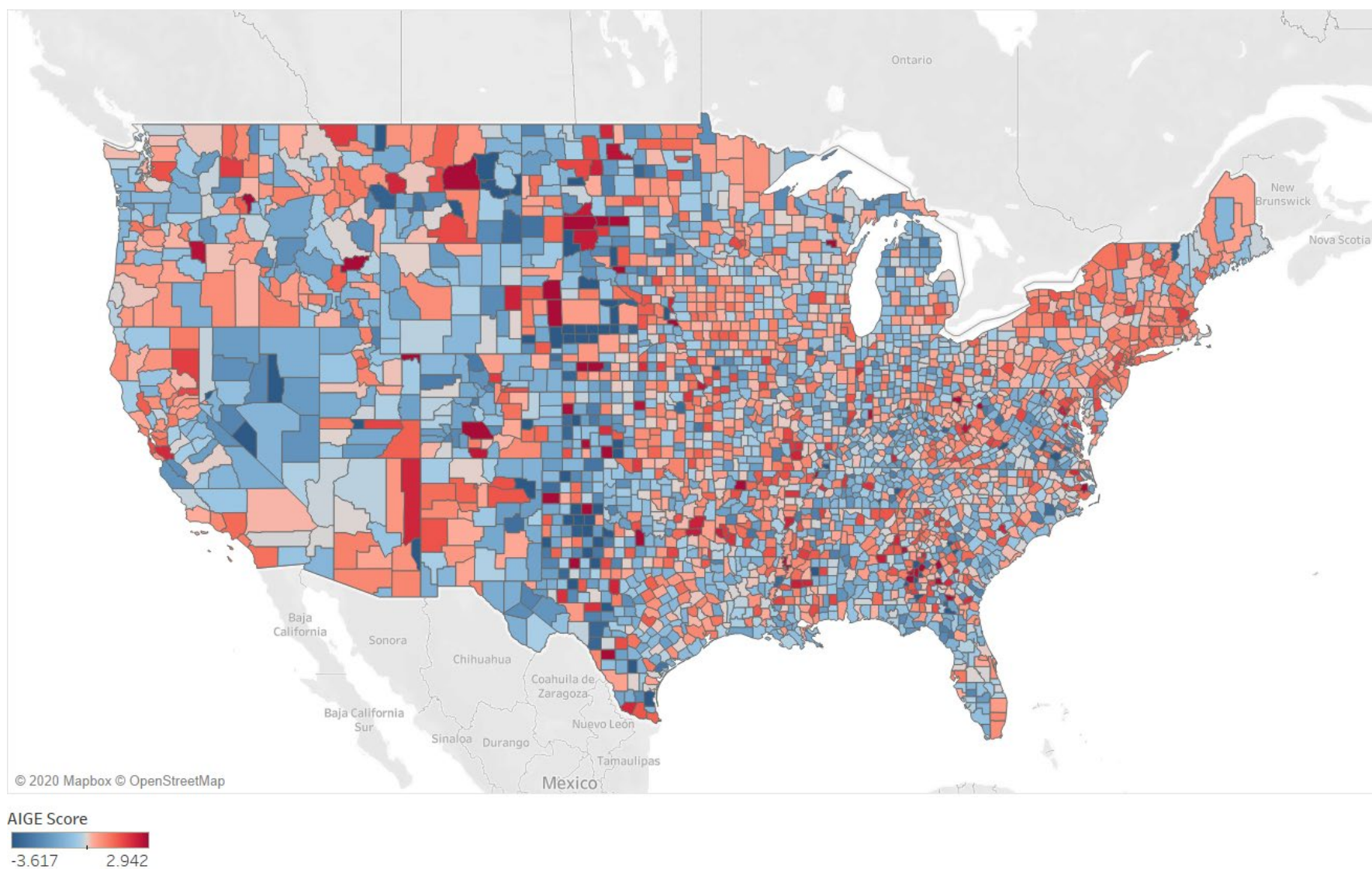
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Figure 1: AIGE in the Continental United States



Note: The charted color represents the county-level AIGE measure winsorized at the 1st and 99th percentile.

Table 1: EFF Application Definitions

AI Application	Definition
Abstract Strategy Games	The ability to play abstract games involving sometimes complex strategy and reasoning ability, such as Chess, Go, or Checkers, at a high level.
Real-Time Video Games	The ability to play a variety of real-time video games of increasing complexity at a high level.
Image Recognition	The determination of what objects are present in a still image.
Visual Question Answering	The recognition of events, relationships, and context from a still image.
Image Generation	The creation of complex images.
Reading Comprehension	The ability to answer simple reasoning questions based on an understanding of text.
Language Modeling	The ability to model, predict, or mimic human language.
Translation	The translation of words or text from one language into another.
Speech Recognition	The recognition of spoken language into text.
Instrumental Track Recognition	The recognition of instrumental musical tracks.

Note: The considered applications are those that have experienced meaningful scientific progress as measured by the Electronic Frontier Foundation at the time of the draft. Definitions are based on those used by the Electronic Frontier Foundation.

Table 2: Occupations with the Highest and Lowest AIOE measures

Rank	Highest Scoring	Lowest Scoring
1	Genetic Counselors	Dancers
2	Financial Examiners	Fitness Trainers and Aerobics Instructors
3	Actuaries	Helpers -- Painters, Paperhangers, Plasterers, and Stucco Masons
4	Purchasing Agents, Except Wholesale, Retail, and Farm Products	Reinforcing Iron and Rebar Workers
5	Budget Analysts	Pressers, Textile, Garment, and Related Materials
6	Judges, Magistrate Judges, and Magistrates	Helpers -- Brickmasons, Blockmasons, Stonemasons, and Tile and Marble Setters
7	Procurement Clerks	Dining Room and Cafeteria Attendants and Bartender Helpers
8	Accountants and Auditors	Fence Erectors
9	Mathematicians	Helpers--Roofers
10	Judicial Law Clerks	Slaughterers and Meat Packers
11	Education Administrators, Postsecondary	Landscaping and Groundskeeping Workers
12	Clinical, Counseling, and School Psychologists	Athletes and Sports Competitors
13	Financial Managers	Fallers
14	Compensation, Benefits, and Job Analysis Specialists	Structural Iron and Steel Workers
15	Credit Authorizers, Checkers, and Clerks	Cement Masons and Concrete Finishers
16	History Teachers, Postsecondary	Terrazzo Workers and Finishers
17	Geographers	Rock Splitters, Quarry
18	Epidemiologists	Plasterers and Stucco Masons
19	Management Analysts	Brickmasons and Blockmasons
20	Arbitrators, Mediators, and Conciliators	Roofers

Note: Occupations are ranked by their constructed AIOE measure at the six-digit Standard Occupational Classification (SOC) level. See draft for a detailed description of the construction of the AIOE measure. Occupation titles are taken from the O*NET database. Highest-scoring occupations are ranked in descending order based on the AIOE measure. Lowest-scoring occupations are ranked in ascending order based on the AIOE measure.

Table 3: Industries with the Highest and Lowest AIIE Measures

Rank	Highest Scoring	Lowest Scoring
1	Securities, Commodity Contracts, and Other Financial Investments and Related Activities	Support Activities for Crop Production
2	Accounting, Tax Preparation, Bookkeeping, and Payroll Services	Services to Buildings and Dwellings
3	Insurance and Employee Benefit Funds	Foundation, Structure, and Building Exterior Contractors
4	Legal Services	Animal Slaughtering and Processing
5	Agencies, Brokerages, and Other Insurance Related Activities	Building Finishing Contractors
6	Nondepository Credit Intermediation	Warehousing and Storage
7	Other Investment Pools and Funds	Fiber, Yarn, and Thread Mills
8	Insurance Carriers	Support Activities for Rail Transportation
9	Software Publishers	Sawmills and Wood Preservation
10	Lessors of Nonfinancial Intangible Assets (except Copyrighted Works)	Support Activities for Water Transportation
11	Agents and Managers for Artists, Athletes, Entertainers, and Other Public Figures	Logging
12	Credit Intermediation and Related Activities (5221 And 5223 only)	Other Specialty Trade Contractors
13	Computer Systems Design and Related Services	Waste Collection
14	Management, Scientific, and Technical Consulting Services	Postal Service (federal government)
15	Monetary Authorities-Central Bank	Highway, Street, and Bridge Construction
16	Office Administrative Services	Truck Transportation
17	Other Information Services	Apparel Knitting Mills
18	Data Processing, Hosting, and Related Services	Seafood Product Preparation and Packaging
19	Business Schools and Computer and Management Training	Local Messengers and Local Delivery
20	Grantmaking and Giving Services	Utility System Construction

Note: Industries are ranked by their constructed AIIE Measure at the four-digit North American Industry Classification System (NAICS) level. See draft for a detailed description of the construction of the AIIE Measure. Industry titles are taken from the Bureau of Labor Statistics. Highest-scoring occupations are ranked in descending order based on the AIIE Measure. Lowest-scoring occupations are ranked in ascending order based on the AIIE Measure.

Online Appendix for
Occupational, Industry, and Geographic Exposure to Artificial Intelligence:
A Novel Dataset and Its Potential Uses

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Appendix A: Construction of the Application-Ability Relatedness Matrix

We link AI applications with the set of abilities in the O*NET database using approximately 2,000 survey responses from “gig workers” from Amazon’s Mechanical Turk (mTurk) web service. mTurk functions as a crowdsourced internet marketplace that facilitates payment for the completion of tasks by workers. Crowdsourcing platforms such as mTurk have increasingly been used to construct datasets for academic research in the field of information systems and management science (e.g., Brynjolfsson et al. 2018, Kim and Luca 2019). Although the use of these tools may raise concerns about external validity, prior work suggests that surveys and experiments executed through online labor market platforms, such as mTurk, are largely generalizable to in-person or laboratory settings (Horton et al. 2011).¹ We rely on these mTurk survey responses to construct a matrix connecting the AI applications to the O*NET occupation-level ability data.

To link the AI applications to workplace abilities, for each of the AI applications considered in our analysis (abstract strategy games, real-time video games, image recognition, visual question answering, image generation, reading comprehension, language modeling, translation, and speech recognition) we survey a sample of approximately 2,000 mTurkers residing in the United States (we solicited answers from approximately 200 respondents for each of the ten AI applications). Each respondent is asked to consider how the AI application is related to each of the 52 abilities considered by O*NET in its occupational definitions. The survey provides descriptions of the AI application from the EFF as well as definitions of the O*NET abilities; for each ability we ask respondents to answer “Yes” or “No” based on whether they believe that the application is related to or could be used for each ability. We opt for a

¹ We also test the sensitivity of the survey responses to using samples based on education-level or subject matter expertise as well as a focus group of computer scientists in Section 3.3.1.

simple “Yes or No” question for simplicity and to limit the extent to which participants can express their own biases about the relationship between AI and the ability because our measure is designed to be agnostic as to whether AI is a complement to or a substitute for labor.

We acknowledge that this crowdsourced approach has weaknesses. Respondents’ perceptions of relatedness may be affected by observed progress on different applications thus far, as well as by lay understandings of how AI technologies can be used in the workplace. Accordingly, they may not recognize or anticipate creative or unconventional ways in which different AI applications might affect certain abilities. Nevertheless, we believe that the responses do provide us with a means to measure the relationship between different AI applications and occupational abilities. An example of a question used for “image recognition” is presented in *Appendix Figure A1*.

Respondents receive \$1 upon completion of the survey, which is designed to take five to ten minutes to complete. The questions are identical for each application-ability combination except for the definitions presented. The order of the abilities presented to each respondent is randomized, and an attention-check question is included in the middle portion of the survey. Respondents that fail the attention check or do not complete the survey entirely are removed from the sample. In addition, using the mTurk platform, we collect background information on respondents regarding the highest level of education achieved and the academic field in which they obtained their highest degree.

We operationalize the survey data by coding “Yes” or “No” responses as a binary variable. Across each application-ability combination, we average the binary measure to construct a measure of how related the AI applications are to the ability. We organize this

measure of application-ability relatedness into a matrix that connects the ten separate EFF AI applications to the 52 O*NET occupational abilities.

Figure A1: mTurk Survey Question

Artificial intelligence in terms of image recognition is defined as the determination of what objects are present in a still picture. For the following human abilities defined below, please answer “Yes” or “No” depending on whether you believe that image recognition by a computer or machine is related to or could be used for each ability.

Peripheral Vision: The ability to see objects or movement of objects to one's side when the eyes are looking ahead.

Do you believe that image recognition by a computer or machine could be used for peripheral vision?

- ☐ Yes
- ☐ No

Appendix B: Discussion of the AIOE Scores for Truck Drivers and Taxi Drivers

Media, policymakers, and researchers appear to be particularly concerned about some occupations that may be negatively affected by advances in AI. For example, AI technologies have shown the potential to outperform radiologists in detecting some abnormalities in x-rays, which has led to much debate over how AI may affect radiology as an occupation in the future (e.g., Agrawal et al. 2019, Hosny et al. 2018, Jha and Topol 2016, Pakdemirli 2019). The O*NET database used in the construction of the AIOE measures does not contain data on workplace abilities required by radiologists specifically;² instead we focus on two other occupations that are also discussed frequently in the context of AI — truck drivers and taxi drivers (e.g., Brynjolfsson et al. 2017, Mochizuki 2019, O’Brien 2019, Vanian 2017). AI technologies in the trucking and taxi industries offer large potential benefits in the form of risk and cost reduction (Brynjolfsson et al. 2017), and advances in AI-based autonomous vehicle technology have led some to believe that these occupations are likely to be disrupted as advances in AI continue (Murphy 2017, Rea et al. 2017).

Despite these concerns, long-haul truck drivers and taxi drivers do not receive particularly high AIOE measures. Long-haul truck drivers have an AIOE measure at the 14th percentile in relation to all other occupations in our sample, while taxi drivers and chauffeurs are at the 34th percentile. In some respects, this may seem surprising. Both occupations require some abilities that are highly exposed to AI according to our methodology, such as problem sensitivity and control precision, and advances in self-driving car technologies may lead some to believe that these occupations are particularly susceptible to disruption. The relatively low AIOE

² The occupations of radiation therapist and radiation technologist are related but require different abilities and entail performing quite different tasks.

measures for the occupations may be driven in part by our decision to focus on the exposure to just AI technologies instead of robotics, sensors, and other technologies.

While the AIOE captures the exposure of occupational abilities and tasks to AI technologies, by design it does not capture advances in robotics technologies. Autonomous vehicles integrate AI technologies, but the driving component itself relies on robotics. Thus, although the AIOE can capture how certain abilities used by truck and taxi drivers are exposed to advances in AI (e.g., the cognitive abilities required to construct a route, the psychomotor abilities involved in precision steering, or the sensory abilities required in depth perception), it does not capture how the interaction between advanced AI and robotics technologies relates to abilities or occupations (e.g., the transition from the AI to the physical actions taken to actually turn a steering wheel and drive a vehicle). Given the large role that physical manipulation and robotics technologies currently have in these occupations, the AIOE suggests that AI itself is unlikely to have an outsized influence on these occupations.

In comparing the two occupations, one may then wonder why taxi drivers are considered to have higher exposure to AI than the long-haul truck drivers. As with the example of surgeons and slaughterhouse workers, the difference is accounted for by the abilities that do not overlap between the two occupations. At least as of now, O*NET suggests that taxi drivers rely more heavily on a number of cognitive and sensory skills that AI may be related to, such as oral comprehension or time sharing (i.e., the ability to shift back and forth between two or more activities or sources of information). Because such abilities are more heavily weighted for taxi drivers than for long-haul truckers, taxi drivers are considered relatively more exposed to AI.

This raises an important facet of our measure as constructed: it is reliant on accurate depictions of the abilities needed within an occupation and does not account for potential

changes in occupational abilities based on advances in technologies. It is possible that, in the future, technological advances will alter whether and how certain abilities are used within an occupation. For example, perhaps the cognitive abilities considered important for taxi drivers now, such as oral comprehension and speech recognition, might become less important in the future if individuals are able to enter their destination using ride-sharing applications or if traditional taxis are replaced by self-driving cars. If that is the case, we may over- or underestimate the relative occupational exposure to AI based on outdated information regarding occupational abilities. However, this approach lets us avoid making assumptions about whether or how AI changes the required abilities or task composition of an occupation. Further, our methodology allows for the AIOE measure to be updated as O*NET releases new updates, providing us with an opportunity to reconstruct measures of AI exposure as the composition of abilities within occupations shifts.

Appendix C: Quantitative Validation of the AIOE and Related Measures

In addition to our qualitative review of occupations and their AIOE measures, we test the robustness of our AIOE measure and seek to validate the measure quantitatively. To do so, we test the robustness of our measure along four separate dimensions: (1) we consider the construction of our matrix linking AI applications to occupational abilities, (2) we test the sensitivity of the measure to which set of AI applications is included, (3) we use job postings to measure the correlation between the AIOE measure and the use of AI skills within an occupation, and (4) we examine how changes in occupational definitions by O*NET affect the AIOE measure over time.

C.1 Robustness of the Application-Ability Matrix

One key component of our methodology is the matrix constructed using mTurk survey data to connect progress in the ten EFF AI applications with the 52 O*NET abilities. In the process of conducting the surveys to construct this matrix, we collect Amazon-provided data on the educational backgrounds of survey respondents. To determine whether our application-ability relatedness scores might be biased by including laypeople without expertise in the phenomenon, we compare the matrix used in our analysis to matrices constructed using samples of only those who completed a graduate degree program and only those whose highest degree was in computer science or engineering. The correlations across the different samples are broken down by EFF AI application in *Appendix Table C1* and by O*NET ability in *Appendix Table C2*.

We find a high degree of correspondence across the matrices, as shown by the correlations between the scores of the survey responses for each EFF AI application and O*NET ability for the different subsamples (we find that the median correlations between our full-sample application-ability relatedness scores and the corresponding scores are 0.98 and 0.94 at the AI

application level and 0.97 and 0.92 at the occupation ability level for the graduate degree and computer science and engineering subsamples respectively). These findings suggest that the AIOE measures are unlikely to be sensitive to using subsamples of highly educated individuals or individuals with a background in computer science or engineering who are likely to have greater understanding of AI technologies.

To further explore this, we constructed an alternative version of our application-ability matrix with the help of a focus group of four computer scientists who conduct research on AI and digital technologies. Although the focus group is small, it consists of subject matter experts who we can expect to have a good understanding of which abilities may be more or less affected by different applications of AI. We use this alternative matrix to construct a version of the AIOE and compare it with the AIOE obtained using the mTurk-constructed matrix. The AIOE measures obtained using the expert-constructed matrix have a strong, positive relationship with the AIOE measures obtained using the mTurk-constructed matrix ($\rho = 0.987, p < 0.001$), suggesting that our matrix linking AI applications to occupational abilities is unlikely to be biased by its reliance on a broad sample of non-experts.

C.2 Sensitivity to Application Selection

In the construction of the AIOE measure, we focus on ten applications of AI for which the EFF has recorded scientific activity and progress from 2010 onward. These applications are consistent with the categorizations used by other academic institutions and organizations (Association for the Advancement of Artificial Intelligence (AAAI) 2021, Papers with Code 2021, Perrault et al. 2019), private firms such as the McKinsey, Deloitte, and Forrester Research, as well as scholars (Bessen et al. 2018, Martínez-Plumed et al. 2020), and our conversations with subject matter experts lead us to believe that they represent fundamental applications of AI that

are likely to cover the most common and likely uses of AI (at least in the medium run). However, there are reasons to be concerned that using this select set of applications could bias our results. Our methodology treats each application of AI as independent since accounting for the various interdependencies would be intractable. Accounting for the relationship between different applications of AI represents a significant challenge as it is often difficult to draw clear lines between applications that reflect similar tasks. For example, one could argue that image recognition is more similar to visual question answering than to the ability to play abstract strategy games. In that sense, the selection of applications might bias our results if we were to “double count” by including applications that have overlapping uses or effects on occupational abilities. In addition, if we exclude an application that is related to a unique set of abilities relative to our considered applications, we may also neglect important ways in which occupations can be exposed to AI.

To address this issue, we conduct two robustness checks that examines the sensitivity of the AIOE measure to excluding certain applications in the construction of the measure. These checks should allow us to understand how sensitive the measure is to the selection of applications.

As a first test, we construct ten alternative versions of the AIOE, each version excluding one of the ten applications of AI we consider in our baseline AIOE measure. We then examine the correlation across these different AIOE measures; the results of this analysis are presented in *Appendix Table C3*.

Across the different measures of the AIOE constructed using different samples of AI applications, we find a strikingly high correlation. Correlations between the baseline AIOE and

the alternative measures constructed to each exclude an application of AI are all greater than 0.99 (and all significant at $p < 0.001$).

As a second check, we construct alternative versions of the AIOE using the set of applications from the EFF that are most frequently referred to by other scholars and experts in AI. We first compile categorizations used by professional and industry organizations, such as the McKinsey Global Institute (Bughin et al. 2017), Forrester Research (Sridharan et al. 2020), and Deloitte (van de Gevel et al. 2017), as well as those constructed by academics, such as Bessen et al. (2018) and Martínez-Plumed et al. (2020), and academic institutions and organizations, such as Stanford's AI Index (Perrault et al. 2019), the Association for the Advancement of Artificial Intelligence (Association for the Advancement of Artificial Intelligence (AAAI) 2021), and the Papers with Code, an organization that has created a repository of research on the state-of-the-art in machine learning (Papers with Code 2021).³ While the categorizations vary across sources, we find a high degree of correspondence across these different sources. Of the ten applications considered by the EFF, five are present in some form across the majority of considered sources: image recognition, language modeling, reading comprehension, speech recognition, and translation. Given that these applications appear to be referred to almost universally as fundamental applications of AI, we believe that they should be a good benchmark to reference the AIOE against. We compare an alternative version of the AIOE constructed with this set of applications to the AIOE constructed with the full set of ten AI applications from the EFF and find a strikingly high correlation ($r = 0.996$; $p < 0.001$).⁴

³ For a breakdown of the applications of AI discussed in each of these sources, please see *Appendix Table C4*.

⁴ Versions of the AIOE constructed either excluding language modeling and reading comprehension or including only one of them have a correlation with the baseline AIOE ranging from 0.997 to 0.999.

The high level of correlation across versions of the AIOE constructed using different samples of applications is due to the similarity in abilities exposed to AI across applications. As discussed in Sections 3.1 and 3.2 above, we find that our measure of AI exposure is highly related to the presence of cognitive and sensory abilities. While each AI application may be more or less related to certain individual abilities, we find that all AI applications are most related to cognitive or sensory abilities, and across the O*NET data, such families of abilities are often used together within an occupation. Accordingly, occupations that rely heavily on such abilities are likely to have high AIOE measures regardless of what sample of AI applications we focus on.

Again, we seem to find suggestive evidence that AI is most likely to affect cognitive and sensory abilities. This makes sense to us given that AI technologies revolve around iterative learning and excel at perception-related tasks (Raj and Seamans 2019). This check provides us with confidence that our results are unlikely to be meaningfully biased based on the sample of AI applications that we consider. Our scores are not sensitive to excluding any of the applications included in our sample and based on this test and the nature of AI technologies, we believe that any applications that we have possibly excluded are likely to also be related to a similar set of cognitive and sensory abilities.

C.3 Relationship between the AIOE and AI Skills in Job Postings

As an additional validation exercise, we evaluate the relationship between the AIOE measure and job skills using Labor Insight data from Burning Glass Technologies. The proprietary Labor Insight data from Burning Glass provide labor market information by compiling job postings from more than 40,000 publicly available sources, such as internet job boards and corporate websites. Burning Glass uses AI technology to analyze the job postings and

provide insight into labor market patterns. The data classify each job posting using the O*NET occupational classifications and provide a list and taxonomy of the skills required for each job posting. Using this information in conjunction with machine learning analysis and qualitative research, Burning Glass classifies over 17,000 unique skills into a taxonomy of skill clusters by grouping together similar skills that are substitutable in many labor market contexts or are often taught or acquired together (Restuccia 2019).

We use the Burning Glass data to measure the relationship between AI skills — which we define as skills within the Burning Glass skill clusters of Artificial Intelligence, Machine Learning, and Natural Language Processing — in an occupation and the AIOE measure.⁵ These clusters consist of skills that require the use of such technologies (rather than focusing on the development of the technologies) and are classified using Burning Glass’s machine learning-based text analysis of job postings.⁶ Presumably, if the AIOE accurately measures occupations with more exposure to AI, we should find a positive correlation between the measure and the count and prevalence of AI-related skills in an occupation. We use data from the most recent version of the Burning Glass data we have access to, covering job postings for the first seven months of 2018. For each occupation, we calculate the average count of these skills required in a job posting, the average percentage of skills listed that are AI skills, and the proportion of job postings that require at least one AI skill. We find a clear positive relationship between the AIOE and the average count of AI skills required in a job posting ($r = 0.263$; $p < 0.001$), the average percentage of AI skills within a job posting ($r = 0.234$; $p < 0.001$), and the proportion of

⁵ Results are robust to only including skills within the Artificial Intelligence skill cluster.

⁶ For example, skills within these three clusters include broad skills such as “Neural Networks,” “Random Forests,” and “Deep Learning,” as well as familiarity with more specific AI tools and techniques such as “AI ChatBot,” “IBM Watson,” “Word2Vec,” and “Vowpal.”

occupational postings that require at least one AI skill ($r = 0.297$; $p < 0.001$).⁷ These patterns are displayed graphically in *Appendix Figure A2*. In *Appendix Figures A3* and *A4*, we replicate the analysis by comparing our AIIE and AIGE measures to AI skills requirements in job postings at the industry and regional levels, respectively. The results are in line with those in *Appendix Figure A1* and show a significant and positive relationship between the AIIE and AIGE measures and AI skills within the industry or geography.

To further corroborate the correlation, we conduct regression analyses to identify the relationship between the AIOE and the presence of AI skills within an occupation. We regress our three measures of AI skills within an occupation on the AIOE controlling for the average count of required skills in an occupation and the average proportion of specialized skills required for an occupation. The results of this analysis, presented in *Appendix Table C5*, demonstrate that even after controlling for such factors, we find statistically significant relationships between the AIOE and the presence of occupational AI skills using both ordinary least squares (OLS) and Poisson models. As an additional robustness check, we conduct this analysis at the job posting-level as well using the most recent version of the Burning Glass data we have access to, covering job postings for the first seven months of 2018. We regress the count of AI skills in a posting, the proportion of AI skills in a posting, and a binary indicator of whether or not the posting includes at least one AI skill on the AIOE controlling for the count of required skills in a posting and the proportion of specialized skills within the posting. Again, using both OLS and Poisson models and either county and sector or state-by-sector fixed effects, we find consistent evidence of a statistically meaningful and positive relationship between the AIOE and the presence of AI skills in an occupation (presented in *Appendix Table C6*).

⁷ These correlations are presented graphically in *Appendix Figure A2*.

*C.4 Relationship between the AIOE and O*NET definitions over time*

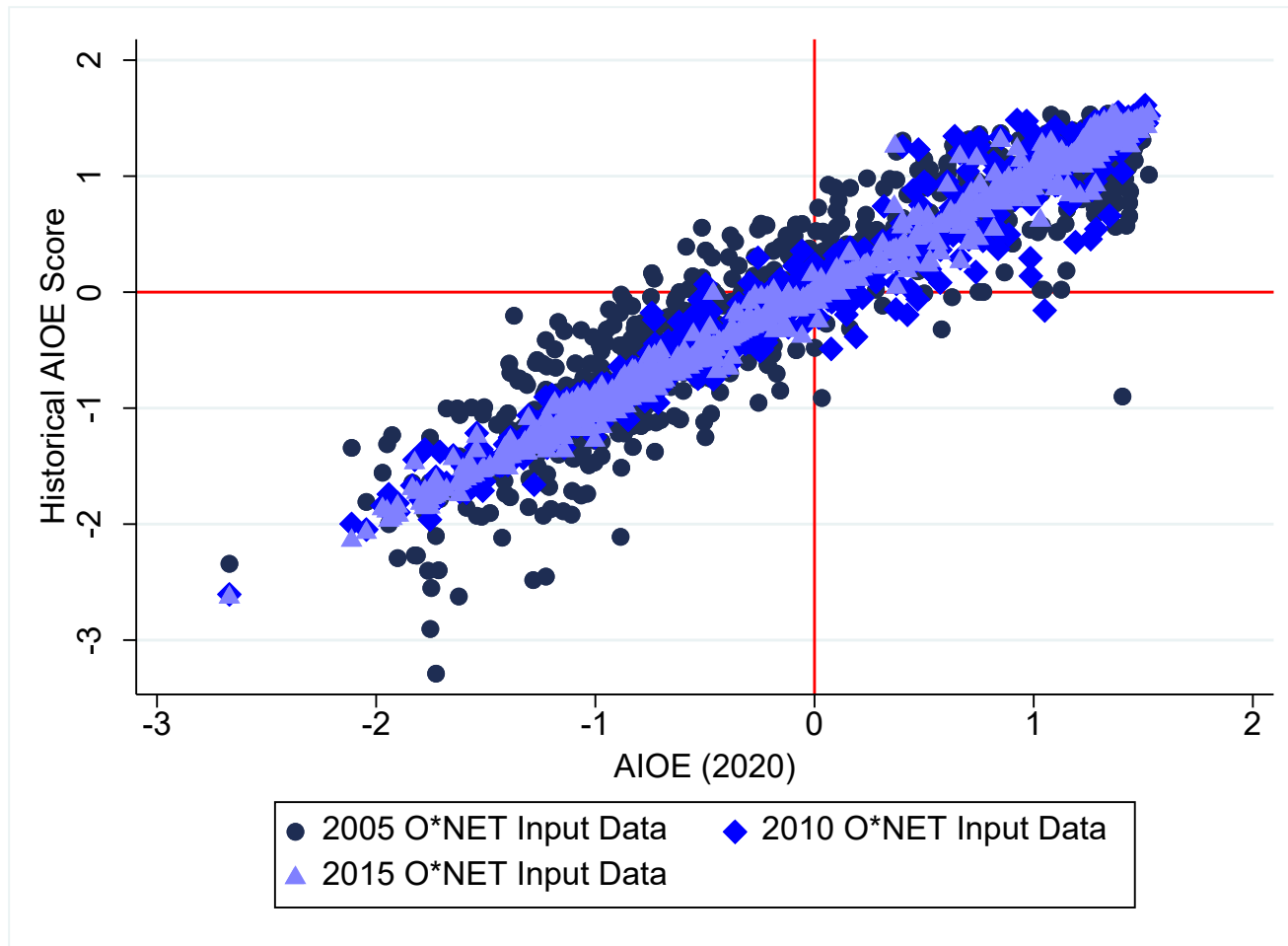
As noted above, the AIOE measure is heavily reliant on accurate depictions of the abilities needed within an occupation and does not account for potential changes in occupational abilities based on advances in technologies. As occupations change over time, O*NET updates its set of occupational definitions and required abilities. To see how durable the AIOE is over time, we examine the sensitivity of the AIOE to using older versions of O*NET's occupational definitions. This test allows us to understand how quickly and to what extent changes in occupational definitions over time affect AIOE measures. Because the AIOE relies on the O*NET database as of 2020, we conduct this test by constructing historical versions of the AIOE using O*NET data from January 2015, 2010, and 2005. In *Appendix Figure C1*, we visually examine the relationship between the 2020 AIOE (x-axis) and historical versions of the AIOE constructed using O*NET input data from 2005, 2010, and 2015 (y-axis).

Figure 1 shows a strong correlation between the 2020 AIOE measure and all three of the historical versions of the AIOE. Not surprisingly, the strength of this correlation increases over time — the 2020 AIOE has a correlation of 0.911 with the 2005 AIOE, 0.982 with the 2010 AIOE, and 0.995 with the 2015 AIOE (all significant at $p < 0.001$). This test indicates that the AIOE is relatively stable over time and that occupational changes (at least in short- to medium-term time horizons) are unlikely to dramatically affect the AIOE values. We note, as occupations undergo larger changes over longer periods of time, our methodology allows for the AIOE measure to be updated as O*NET releases updates.

It is worth noting however, that using updated O*NET data may bias results due to how the demand for abilities may change for advances in AI that complement vs. substitute for occupational abilities. For abilities complemented by AI, it is likely that O*NET would note a

relative increase in the demand for such abilities within occupations, making these occupations appear even more exposed than before. However, for skills that are substituted, O*NET would note a relative decrease in the demand for such abilities within an occupation, perhaps leading us to conclude that the occupation is less exposed than before. This is a limitation that is likely to exist in any setting that relies on abilities or skills to describe occupations over time. We encourage potential users of the measure to keep this in mind and emphasize that our measure is a forward-looking measure of AI exposure based on the occupational definitions at a given point in time.

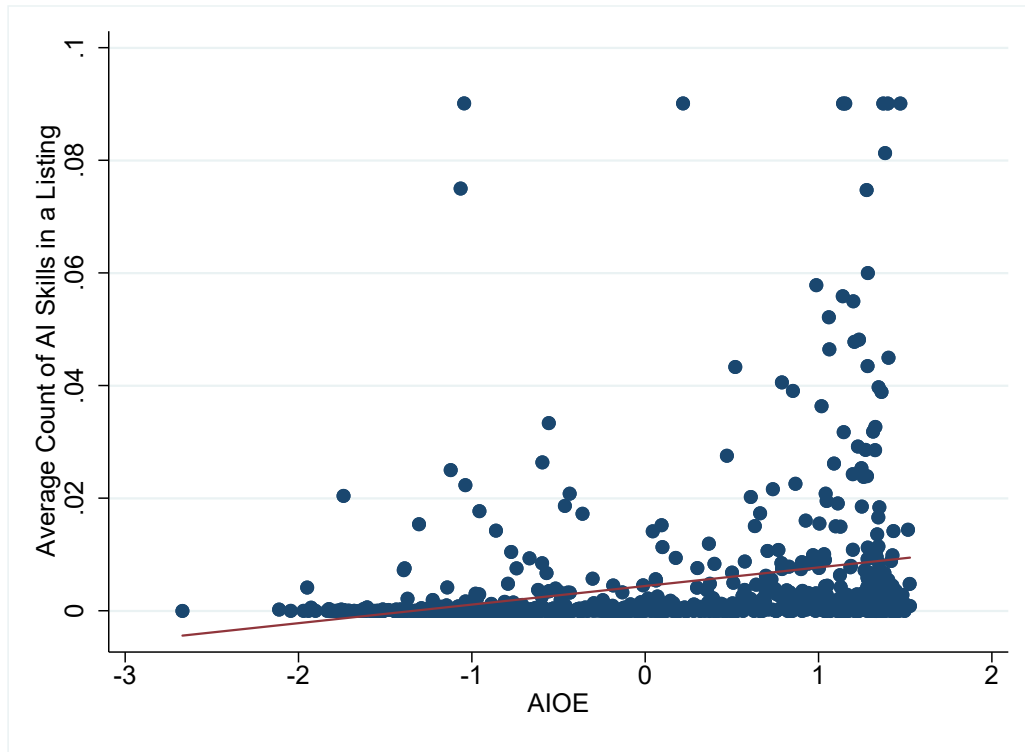
Appendix Figure C1: AIOE with Historical O*NET Input Data



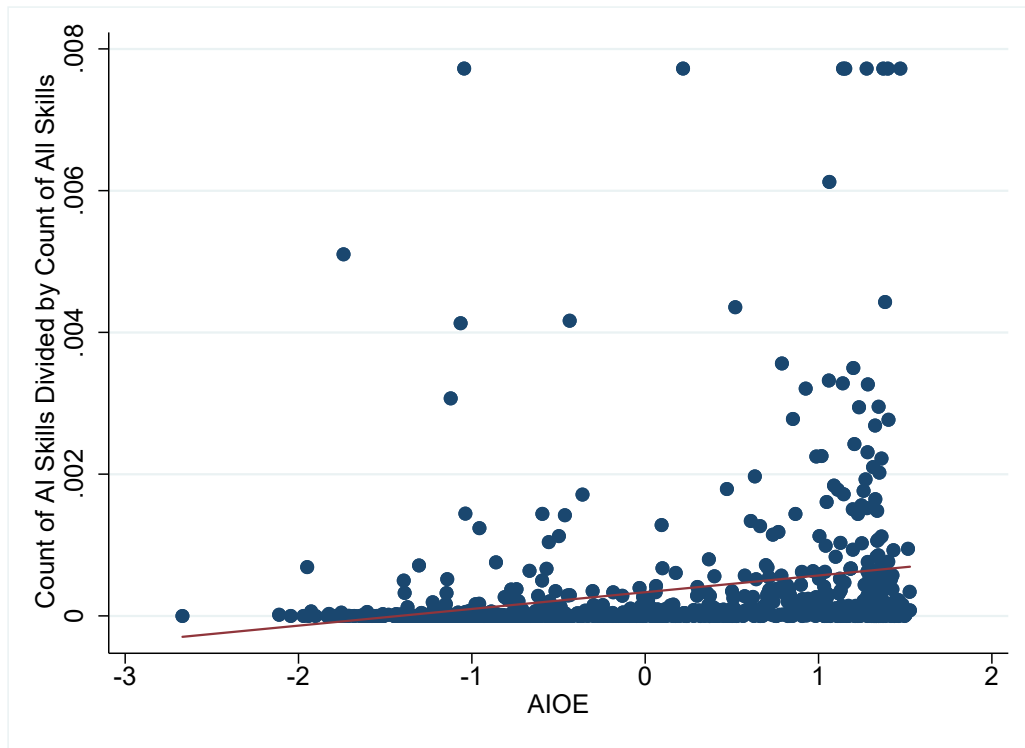
Note: The x-axis displays the standardized AIOE using O*NET data on occupational abilities as of January 2020. The charted series represent historical versions of the AIOE using O*NET occupational ability data as of January 2005, 2010, and 2015. Only occupations that have a 2020 AIOE are included in this figure.

Figure C2: Correlation between AIOE and Occupation AI Skills

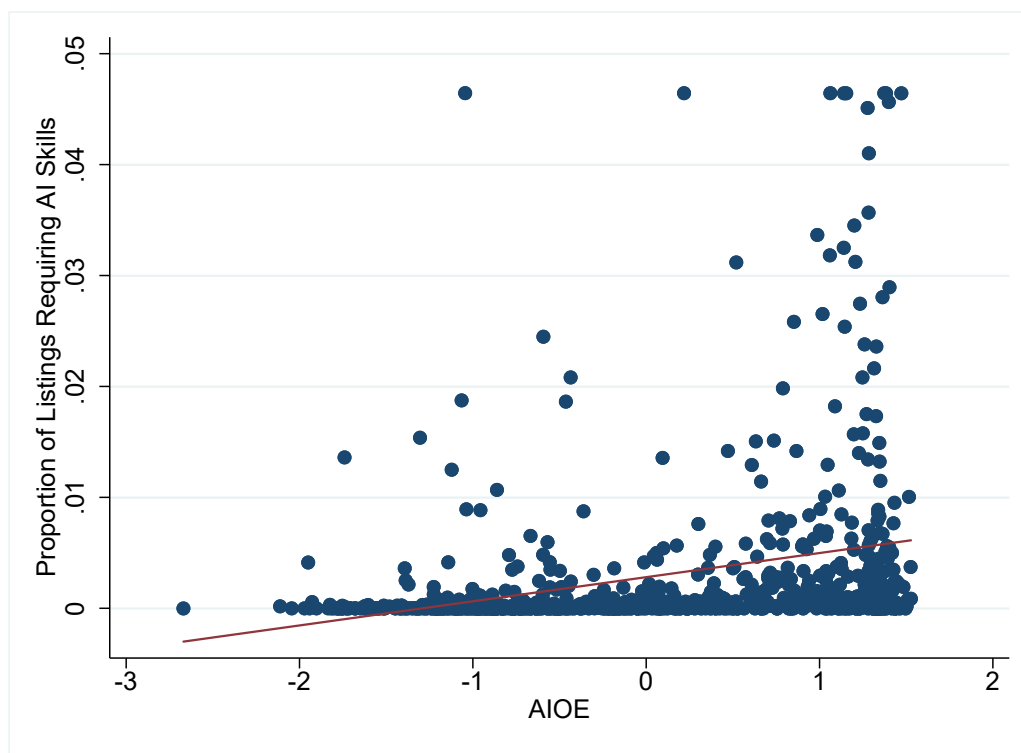
Panel A. AIOE and the Average Count of AI Skills Required in a Job Posting.



Panel B. AIOE and the Average Percentage of AI Skills Required in a Job Posting.



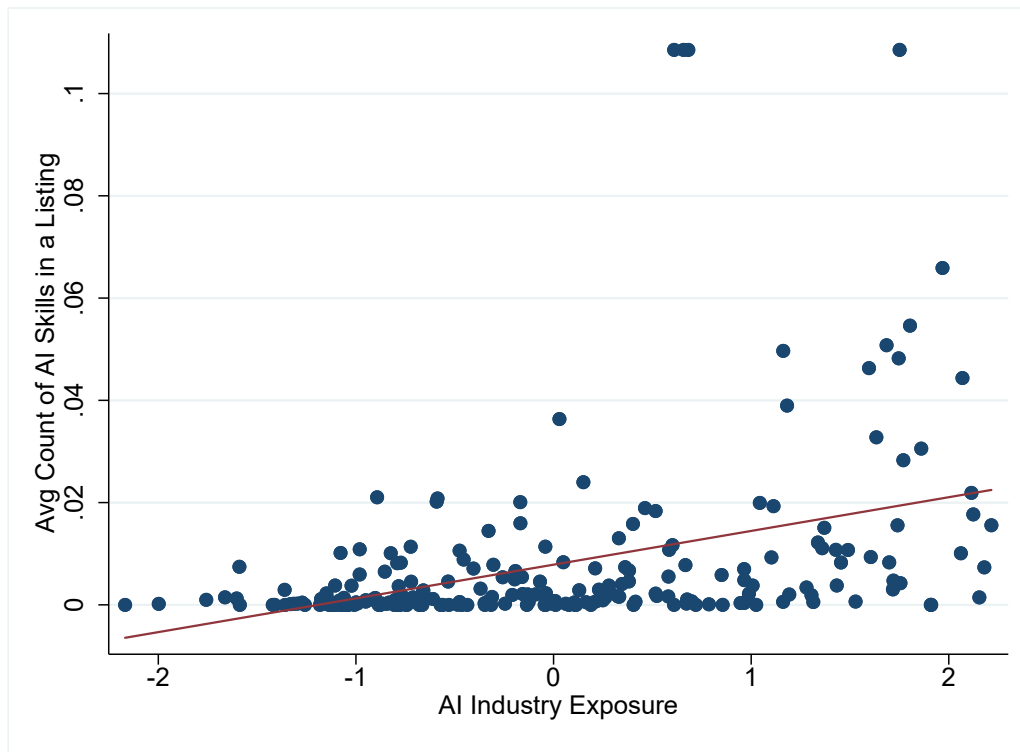
Panel C. AIOE and the Proportion of Job Postings Requiring AI Skills.



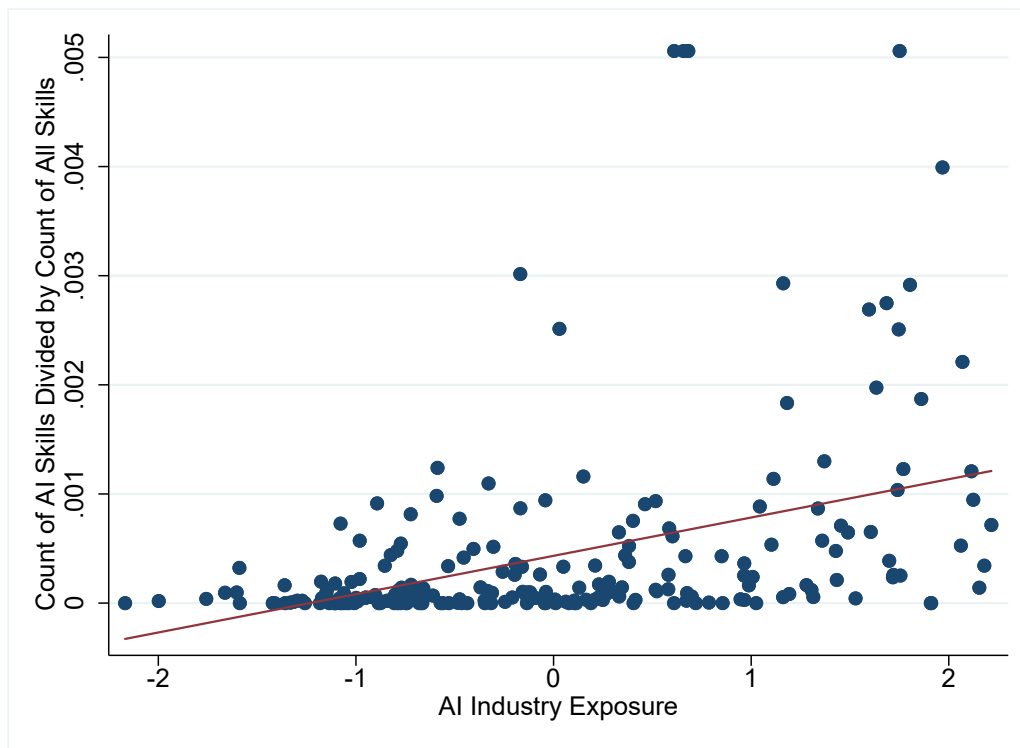
Note: Data on AI skills within occupations come from Burning Glass Technologies Labor Insight data. AI skills include all skills within the artificial intelligence, machine learning, or natural language processing families. Burning Glass provides data on online job postings and uses machine learning technologies to classify required skills. AI skills refers to the use of AI technologies, not to their development. Data are from job postings in 2018 (January through July inclusive), the most recent Burning Glass data accessible to the authors at the time of writing.

Appendix Figure C3: Correlation between AIEE and Industry Occupation AI Skills

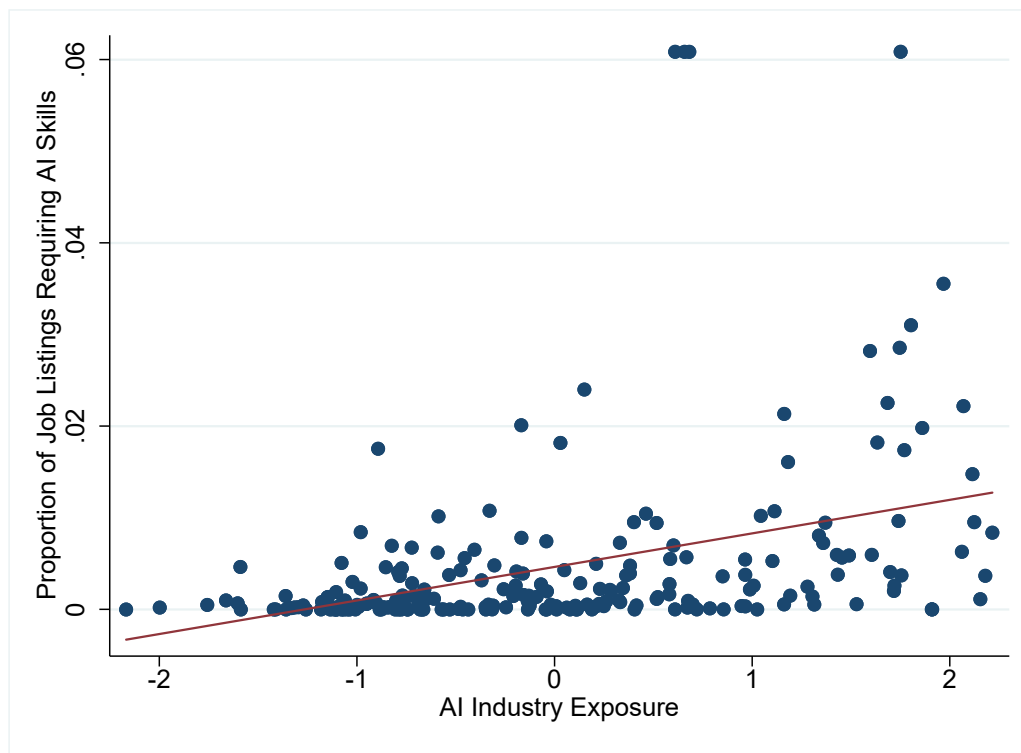
Panel A. AIEE and the Average Count of AI Skills Required for an Industry Posting.



Panel B. AIOE and the Average Percentage of AI Skills Required in an Industry Posting.



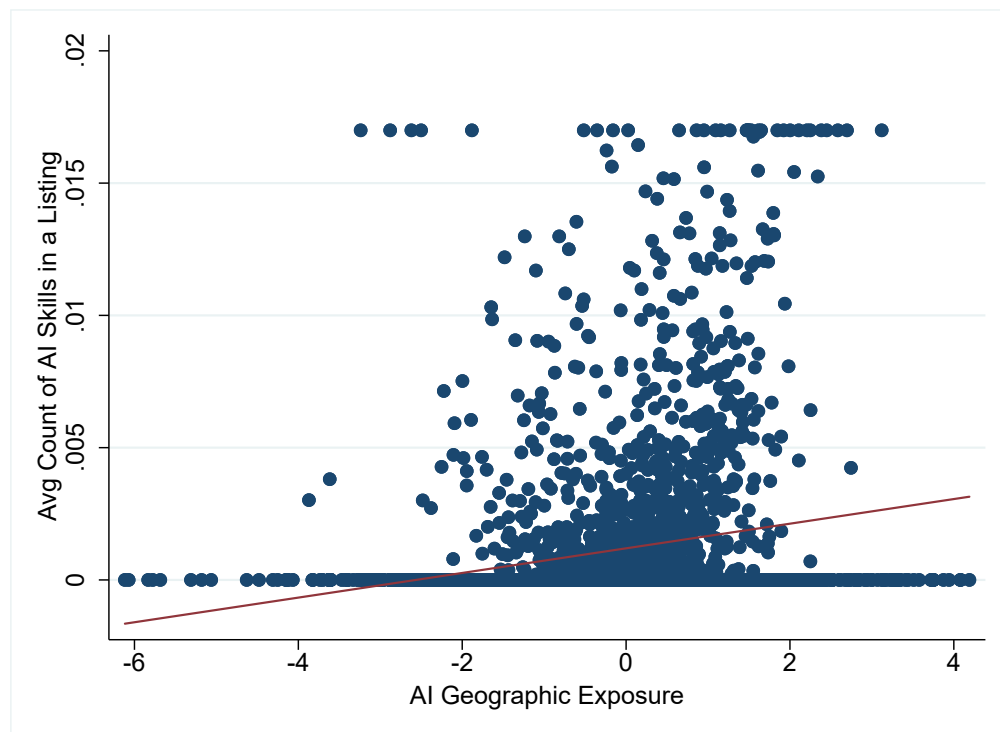
Panel C. AIIE and the Proportion of Industry Postings Requiring AI Skills.



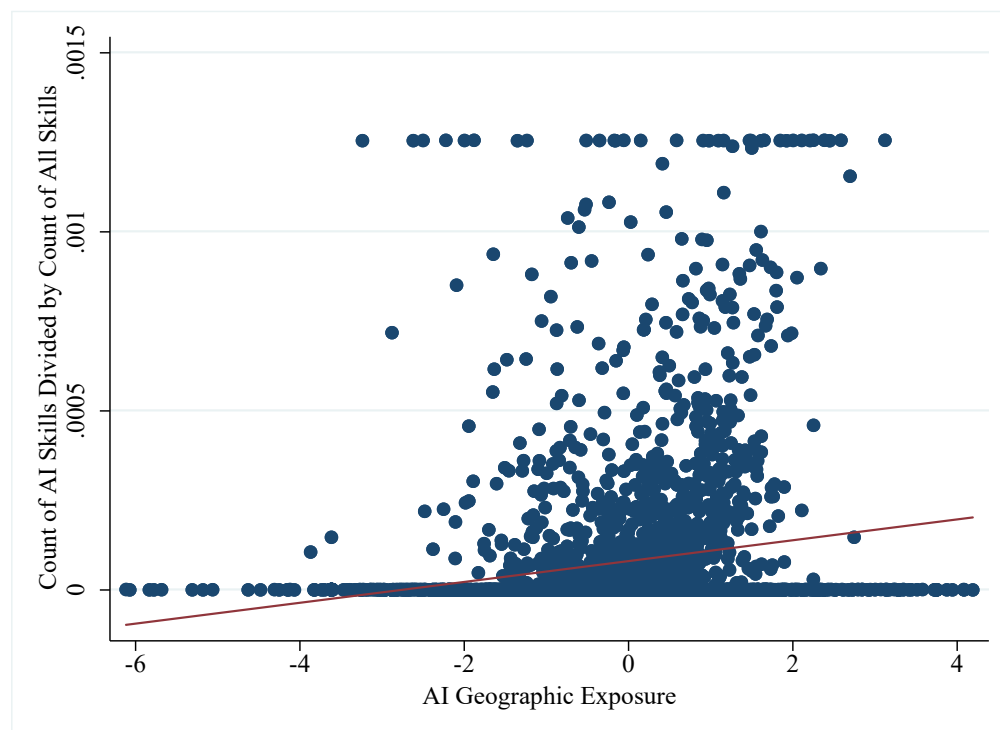
Note: Data on AI skills within occupations come from Burning Glass Technologies Labor Insight data. AI skills include all skills within the artificial intelligence, machine learning, or natural language processing families. Burning Glass provides data on online job postings and uses machine learning technologies to classify required skills. AI skills refers to the use of AI technologies, not to their development. Data are from job postings in 2018 (January through July inclusive), the most recent Burning Glass data accessible to the authors at the time of writing.

Appendix Figure C4: Correlation between AIGE and Within-County Occupation AI Skills

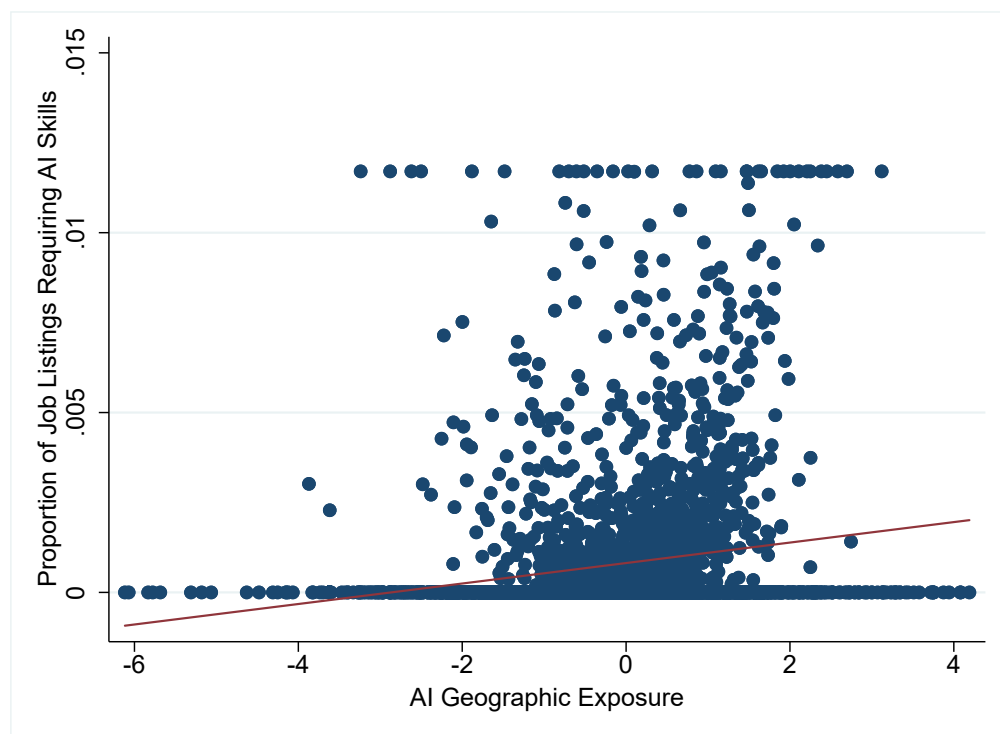
Panel A. AIGE and the Average Count of AI Skills Required for Within-County Postings.



Panel B. AIGE and the Average Percentage of AI Skills Required for Within-County Postings.



Panel C. AIGE and the Proportion of Within-County Postings Requiring AI Skills.



Note: Data on AI skills within occupations come from Burning Glass Technologies Labor Insight data. AI skills include all skills within the artificial intelligence, machine learning, or natural language processing families. Burning Glass provides data on online job postings and uses machine learning technologies to classify required skills. AI Skills refers to the use of AI technologies, not to their development. Data are from job postings in 2018 (January through July inclusive), the most recent Burning Glass data accessible to the authors at the time of writing.

Appendix Table C1: Correlation Coefficients across mTurk Subsamples by EFF AI Application

EFF AI Application	Correlation Coefficient between the Subsample and the Full Sample	
	Graduate Students	Computer Science or Engineering Backgrounds
Abstract Strategy Games	0.97	0.92
Real-Time Video Games	0.96	0.87
Image Recognition	0.97	0.94
Visual Question Answering	0.98	0.90
Generating Images	0.98	0.94
Reading Comprehension	0.99	0.95
Language Modeling	0.99	0.96
Translation	0.99	0.95
Speech Recognition	0.99	0.94
Instrumental Track Recognition	0.98	0.94

Note: Correlation coefficients are calculated for the vectors of responses of the different subsamples listed in the mTurk survey used to construct the matrix connecting AI applications with O*NET abilities.

Appendix Table C2: Correlation Coefficients across mTurk Subsamples by O*NET Ability

O*NET Abilities	Correlation Coefficient between the Subsample and the Full Sample	
	Graduate Students	Computer Science or Engineering Backgrounds
Arm-Hand Steadiness	0.97	0.92
Auditory Attention	0.98	0.93
Category Flexibility	0.87	0.91
Control Precision	0.97	0.98
Deductive Reasoning	0.89	0.94
Depth Perception	0.98	0.99
Dynamic Flexibility	0.91	0.95
Dynamic Strength	0.79	0.86
Explosive Strength	0.87	0.86
Extent Flexibility	0.91	0.91
Far Vision	0.98	0.95
Finger Dexterity	0.97	0.97
Flexibility of Closure	0.92	0.87
Fluency of Ideas	0.92	0.91
Glare Sensitivity	0.99	0.97
Gross Body Coordination	0.94	0.96
Gross Body Equilibrium	0.91	0.85
Hearing Sensitivity	0.99	0.91
Inductive Reasoning	0.79	0.80
Information Ordering	0.85	0.84
Manual Dexterity	0.97	0.98
Mathematical Reasoning	0.97	0.87
Memorization	0.88	0.72
Multilimb Coordination	0.98	0.93
Near Vision	0.99	0.95
Night Vision	0.97	0.96
Number Facility	0.99	0.90
Oral Comprehension	0.99	0.95
Oral Expression	0.96	0.87
Originality	0.91	0.84
Perceptual Speed	0.96	0.91
Peripheral Vision	0.98	0.96
Problem Sensitivity	0.89	0.95
Rate Control	0.98	0.98
Reaction Time	0.97	0.90
Response Orientation	0.98	0.92
Selective Attention	0.96	0.62
Sound Localization	0.97	0.97
Spatial Orientation	0.99	0.97
Speech Clarity	0.96	0.83
Speech Recognition	0.99	0.92
Speed of Closure	0.91	0.91
Speed of Limb Movement	0.97	0.93
Stamina	0.94	0.84
Static Strength	0.89	0.81
Time Sharing	0.95	0.91
Trunk Strength	0.88	0.72
Visual Color Determination	0.99	0.98
Visualization	0.99	0.96
Wrist-Finger Speed	0.98	0.94
Written Comprehension	0.96	0.93
Written Expression	0.98	0.89

Note: Correlation coefficients are calculated for the vectors of responses of the different subsamples listed in the mTurk survey used to construct the matrix connecting AI applications with O*NET abilities.

Appendix Table C3: Selection of AI Applications

		<i>AIOE Excluding...</i>									
	Baseline AIOE measure	Abstract Strategy Games	Real Time Video Games	Image Recognition	Visual Question Answering	Image Generation	Reading Comprehension	Language Modeling	Translation	Speech Recognition	Instrumental Track Recognition
Baseline AIOE measure	1.0000										
<i>AIOE Excluding...</i>											
Abstract Strategy Games	0.9998	1.0000									
Real Time Video Games	0.9996	0.9998	1.0000								
Image Recognition	0.9995	0.9998	0.9999	1.0000							
Visual Question Answering	0.9997	0.9999	0.9999	1.0000	1.0000						
Image Generation	0.9995	0.9998	0.9998	0.9999	0.9999	1.0000					
Reading Comprehension	0.9994	0.9989	0.9981	0.9981	0.9985	0.9983	1.0000				
Language Modeling	0.9989	0.9978	0.9971	0.9969	0.9974	0.9971	0.9996	1.0000			
Translation	0.9990	0.9980	0.9974	0.9972	0.9977	0.9974	0.9997	1.0000	1.0000		
Speech Recognition	0.9995	0.9986	0.9983	0.9980	0.9984	0.9980	0.9995	0.9998	0.9998	1.0000	
Instrumental Track Recognition	0.9998	0.9994	0.9994	0.9992	0.9994	0.9991	0.9989	0.9987	0.9989	0.9995	1.0000

Note: Table presents a correlation matrix displaying the relationship between different variations of the AIOE constructed using different samples of AI applications. The Baseline AIOE measure uses the ten EFF AI applications described in Table 1. The variations are constructed by excluding a single application. All correlations are significant at $p < 0.001$.

Appendix Table C4: Applications of AI Across Sources

Source	Type of Source	AI Categories
Bughin JR, Hazan E, Ramaswamy S, Chui M, Allas T, Peter D, Henke N, Trench M (2017) <i>Artificial Intelligence: The Next Digital Frontier</i> (McKinsey Global Institute).	Industry	Decision management, image recognition and video processing, machine learning and deep learning, natural language generation, speech recognition, virtual agents or artificial conversational entities.
Sridharan S, Leganza G, Vale J (2020) <i>Research Overview: Artificial Intelligence</i> (Forrester Research).	Industry	AI-optimized hardware, biometrics, decision management, deep learning platforms, machine learning platforms, natural language generation, natural language processing, robot process automation, speech recognition, virtual agents.
van de Gevel J, Broersen S, Wolvius C (2017) Applications of Artificial Intelligence. <i>Deloitte</i> . Retrieved (February 4, 2021), https://www2.deloitte.com/se/sv/pages/technology/articles/part3-applications-of-artificial-intelligence.html .	Industry	Image recognition, games, Q&A, speech recognition, translation.
Association for the Advancement of Artificial Intelligence (AAAI) (2021) AITopics. Association for the Advancement of Artificial Intelligence (AAAI). Retrieved (February 25, 2021), https://aitopics.org/search .	Academic	Assistive technologies, challenges, cognitive science, games, history, human-centered computing, issues, machine learning, natural language, representation and reasoning, robots, science fiction, speech, systems and languages, the future, vision.
Bessen JE, Impink SM, Reichensperger L, Seamans R (2018) <i>The Business of AI Startups</i> (Social Science Research Network, Rochester, NY).	Academic	Decision management, natural language classification, natural language generation, natural language translation, natural language understanding and text analysis, robotic process automation, sentiment/emotion analysis, speech recognition, virtual agents/chatbots, visual recognition.
Martínez-Plumed F, Tolan S, Pesole A, Hernández-Orallo J, Fernández-Macías E, Gómez E (2020) Does AI Qualify for the Job?: A Bidirectional Model Mapping Labour and AI Intensities. <i>Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society</i> . (ACM, New York NY USA), 94–100.	Academic	Automated vehicles, computer vision, game playing, information retrieval, machine translation, music analysis, prediction, robotic navigation and interaction, speech recognition, text summarization.
Papers with Code (2021) Browse State-of-the-Art. <i>Papers with Code</i> . Retrieved (February 25, 2021), https://paperswithcode.com/sota .	Academic	Adversarial, audio, computer vision, medical, music, natural language processing, playing games, reasoning, robots, speech.
Perrault CR, Shoham Y, Brynjolfsson E, Clark J, Etchemendy J, Grosz B, Lyons T, Manyika J, Mishra S, Niebles JC (2019) <i>The AI Index 2019 Annual Report</i> (AI Index Steering Committee, Human-Centered AI Institute, Stanford University).	Academic	Activity recognition in video, human-level performance milestones, image classification, image generation, natural language understanding, semantic segmentation, visual question answer.

Note: Table contains sources used to identify various categorizations of applications of AI.

Appendix Table C5: AIOE and Occupation-Level AI Skills

	(1)	(2)	(3)	(4)	(5)	(6)
	Average Count of AI Skills		Average Percent AI Skills		Proportion of Postings Requiring AI Skills	
	OLS	Poisson	OLS	Poisson	OLS	Poisson
AIOE	0.001 (0.000)	0.470 (0.159)	0.010 (0.004)	0.504 (0.185)	0.001 (0.000)	0.525 (0.145)
Average Count of Skills	0.001 (0.000)	0.234 (0.027)	0.009 (0.002)	0.183 (0.033)	0.001 (0.000)	0.231 (0.027)
Average Proportion of Specialized Skills	0.006 (0.003)	1.716 (0.823)	0.049 (0.023)	1.824 (0.876)	0.004 (0.002)	1.714 (0.760)
Observations	772	772	772	772	772	772
R-squared or Pseudo R-squared	0.170	0.100	0.108	0.112	0.197	0.096

Note: This table presents the results of regression analysis estimating the relationship between the AIOE and the presence of AI skills at the occupation-level. The sample includes all occupations with a constructed AIOE measure using O*NET input data from 2020. The Average Count of AI Skills is the average count of skills required for occupational job postings that are classified as AI skills, the Percent AI skills measures the average count of required AI skills divided by the total count of required skills for occupational job postings, and Proportion of Postings Requiring AI Skills is the proportion of occupational job postings that require at least one AI skill. AI skills are defined as skills within the Artificial Intelligence, Machine Learning, or Natural Language Processing skill clusters as defined by Burning Glass Technologies. We control for the average total count of required skills in a job posting and the average proportion of required specialized skills over total required skills as classified by Burning Glass Technologies. Standard errors clustered at the occupation-level are displayed in parentheses.

Appendix Table C6: AIOE and Posting-Level AI Skills

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Count of AI Skills				Percent AI Skills				Any AI Skills in Occupation			
	OLS	Poisson	OLS	Poisson	OLS	Poisson	OLS	Poisson	OLS	Poisson	OLS	Poisson
AIOE	0.0017 (0.0008)	1.2827 (0.2126)	0.0022 (0.0009)	1.3852 (0.2148)	0.0003 (0.0001)	1.2220 (0.2010)	0.0003 (0.0001)	1.3426 (0.2054)	0.0014 (0.0005)	1.2591 (0.1730)	0.0017 (0.0005)	1.3529 (0.1758)
Count of Skills	0.0018 (0.0005)	0.0340 (0.0010)	0.0019 (0.0005)	0.0380 (0.0012)	0.0000 (0.0000)	0.0269 (0.0015)	0.0000 (0.0000)	0.0300 (0.0018)	0.0009 (0.0002)	0.0331 (0.0010)	0.0009 (0.0002)	0.0365 (0.0010)
Proportion Specialized Skills	0.0188 (0.0061)	4.2163 (0.2956)	0.0194 (0.0063)	4.3406 (0.2898)	0.0012 (0.0004)	4.2671 (0.3093)	0.0013 (0.0004)	4.4085 (0.3018)	0.0101 (0.0030)	3.6474 (0.2392)	0.0104 (0.0032)	3.7623 (0.2422)
County FEs?	Yes	Yes	No	No	Yes	Yes	No	No	Yes	Yes	No	No
Sector FEs?	Yes	Yes	No	No	Yes	Yes	No	No	Yes	Yes	No	No
Sector X State FEs?	No	No	Yes	Yes	No	No	Yes	Yes	No	No	Yes	Yes
Observations	14,689,878	13,266,255	14,877,546	14,290,025	14,689,878	13,266,255	14,877,546	14,290,025	14,689,878	13,266,255	14,877,546	14,290,025
R-squared or Pseudo R-squared	0.0169	0.2400	0.0155	0.2317	0.0100	0.1646	0.0081	0.1542	0.0210	0.2092	0.0193	0.2023

Note: This table presents the results of regression analysis estimating the relationship between the AIOE and the presence of AI skills at the job-posting-level. The sample includes all job postings within the Burning Glass data for January through July 2018 (the latest version of the data available to the authors). The Count of AI Skills is the count of skills required for the job posting classified as AI skills, the Percent AI skills measures the count of required AI skills divided by the total count of required skills, and Any AI Skills in Occupation is a binary variable that takes the value of one if the job posting requires any AI skills. AI skills are defined as skills within the Artificial Intelligence, Machine Learning, or Natural Language Processing skill clusters as defined by Burning Glass Technologies. We control for the total count of required skills in a job posting and the proportion of required specialized skills over total required skills as classified by Burning Glass Technologies. Standard errors clustered at the occupation-level are displayed in parentheses.

Appendix D: Comparing the AIOE to Other Existing Datasets

Other researchers have created datasets to measure the exposure to and impact of AI on occupations and industries. In this section, we briefly review them before explaining how our dataset is unique.

D.1 Other Existing Datasets

One notable study attempting to link occupations to AI technologies was undertaken by Frey and Osborne (2017). The authors work with a group of experts to forecast what job tasks may be particularly susceptible to automation. The authors focus particularly on machine learning and its application to mobile robotics and propose a model to predict the extent of computerization's impact on non-routine tasks, noting potential engineering bottlenecks for tasks involving high levels of perception or manipulation, creative intelligence, and social intelligence. After categorizing tasks by their susceptibility to automation, Frey and Osborne map these tasks to occupations, creating a measure for how susceptible to automation each occupation is.

A similar, task-based approach was undertaken by Brynjolfsson, Mitchell, and Rock (2018) and Brynjolfsson et al. (2020). Brynjolfsson and co-authors construct a rubric to ascertain how “suitable for machine learning” different workplace activities are, and then use a crowdsourcing service to apply this rubric to the full range of work activities outlined by the Bureau of Labor Statistics. They aggregate these to the occupation level to quantify the extent to which tasks within an occupation are suitable for machine learning. In addition, Brynjolfsson and co-authors (2020) present an estimate of the annual wage bill susceptible to exposure by two-digit NAICS industry, providing some sense of industry exposure to AI.

Other studies have used patent data to study the relationship between AI and labor. Mann and Püttmann (2017) apply a machine learning algorithm to all US patents granted from 1976 to

2014 to identify patents related to automation (an automation patent is defined as a “device that operates independently from human intervention and fulfills a task with reasonable completion”). They then link the automation patents to the industries and geographies they are likely to be used in and find that, although automation causes manufacturing employment to fall, it increases employment in the service sector and overall has a positive impact on employment. Mann and Püttmann (2017) also use a similar methodology to construct a measure of available automation technology at the three-digit SIC level, and find that the number of automation patents is positively related to investment in computer capital and robot shipments. Webb (2020) similarly uses a machine learning algorithm to identify which tasks are most likely to be automated by AI, and suggests that high-skill tasks and occupations will be particularly affected by advances in AI technology.

Several other reports by private firms and research institutes attempt to identify which industries are more or less affected by advances AI technology. Using Webb’s (2020) occupation-level exposure scores, the Brookings Institution published a report that constructed industry-level exposure to AI automation in which it suggests that the advancement of AI technology is highest among primary and secondary industries such as manufacturing, agriculture, and resource extraction (Muro et al. 2019). The McKinsey Global Institute released a report that estimated the potential value of AI by sector based on internal models of the relative effectiveness of AI in comparison with existing technologies and the potential use cases across industries (Chui et al. 2018). Similarly, Accenture estimated the potential gains to industry profit from advances in AI technology using a methodology that links the probability of automation at the occupation level to improvements in AI over time and to total factor productivity data across industries (Purdy and Daugherty 2017). While these reports provide useful context and are a

valuable resource given the nascent state of the field, they provide limited visibility into the methodology used to calculate the results and produce proprietary data that are difficult to access and therefore of limited use.

D.2 Distinctions between the AIOE and Existing Datasets

The studies outlined above make important contributions to what is a nascent effort to understand the effect of AI on labor and occupations. However, we believe that our measure has unique features and strengths. First, our methodology considers specific applications of AI (e.g., image recognition, speech recognition, and others) and links them to workplace abilities and then to occupations, industries, and geographies. Several of the existing measures consider the effect of automation (Frey and Osborne 2017, Mann and Püttmann 2017) on labor, without specifying whether automation occurs via AI, robots, sensors, or another type of technology. And even measures that specifically consider the effect of AI or machine learning on labor (Brynjolfsson et al. 2018, 2020, Webb 2020) do not discern between specific applications of AI.

Second, our measure identifies the relative *exposure* to AI but remains agnostic as to whether AI complements or substitutes for tasks and labor. In contrast, many of the other measures (e.g., Brynjolfsson et al. 2020, Frey and Osborne 2017, Webb 2020) are focused on the potential for new technologies to substitute for or automate occupations or tasks. To be sure, substitution may well occur in many cases. But we believe it is important to be able to empirically study conditions under which exposure to AI will substitute for or complement labor, especially as we still do not yet know whether or how AI will affect labor markets.

Third, our approach provides a snapshot of occupational exposure to AI based on the current nature of occupations. While other measures use forward-looking approaches that rely on expert projections or crowdsourced evaluations to identify which tasks may be suitable to AI or

automation or how the task composition of an occupation may change as it is affected by AI (Brynjolfsson et al. 2018, 2020, Frey and Osborne 2017), we consider the task composition of an occupation as fixed in our analysis based on the O*NET occupation data. While this prevents us from predicting how task composition and skill content of occupations change over time, it provides us with an accurate representation of AI exposure in the present and allows us to update our measure moving forward as the abilities and skills required within occupations change. Further, it allows us to avoid making assumptions about how AI will affect different occupational tasks (consistent with our agnostic approach to whether AI substitutes for labor).

These different approaches are complementary and can be useful to different researchers depending on the specific questions being asked. Below we discuss in more detail potential applications of the AIOE, AIIE, and AIGE measures.